The Consumer Cost of Pay-for-Delay: Evidence from the U.S. Testosterone Market*

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Abstract

This paper quantifies consumer harm from the Pay-for-Delay agreement that delayed generic entry into the U.S. Testosterone Replacement Therapy market from 2006 until 2015. I estimate a two-level nested logit demand model on monthly product-level data, recover marginal costs from Bertrand-Nash first-order conditions, and simulate counterfactual earlier generic entry by augmenting the choice set with fictitious generics and solving for equilibrium prices. Earlier entry would have made generics available at prices 43–50% below branded levels. Total consumer damages amount to $6,451 million, a 43.6% consumer surplus increase. The supply-side analysis shows that the agreement was highly profitable for the branded manufacturer.

Keywords: Structural model, pharmaceutical competition, antitrust, market power, pay-for-delay, drug entry, patent litigation.
JEL codes: L13, L41, I11, K21.

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1 Introduction

Generic competition is the main mechanism through which drug prices fall once patent protection ends. In the United States, the Hatch-Waxman Act allows a generic manufacturer to challenge a brand-name patent before it expires and, if the challenge succeeds, to enter the market years before the patent expires. Generic entry then reduces prices sharply in the generic segment (Grabowski and Vernon, 1992; Frank and Salkever, 1997; Conrad and Lutter, 2019). However, brand-name firms can settle the patent litigation instead of fighting it. In a *pay-for-delay* (PFD) agreement, also known as a reverse payment settlement, the brand pays the generic challenger to withdraw its challenge and postpone its entry (Hemphill, 2006). In this way, the two firms share the monopoly rent, while consumers continue to pay the monopoly price.

Whether these settlements harm consumers, and by how much, is a question that courts must now answer case by case. In *FTC v. Actavis* (2013) the U.S. Supreme Court ruled that PFD agreements are not illegal *per se* but must be evaluated under the rule of reason, weighing their anticompetitive effects against their possible justifications (Federal Trade Commission, 2013). The Federal Trade Commission (FTC) estimates that these agreements cost American consumers about $3.5 billion per year (Federal Trade Commission, 2019). Yet aggregate figures are of limited use in a rule-of-reason proceeding, which requires an estimate of the harm caused by one specific agreement in one specific market. Producing such an estimate is difficult for two reasons. First, the relevant counterfactual (a market in which generic entry happened earlier) is never observed. Second, the price responses of the incumbents to that entry cannot be measured, so they have to be inferred.

This paper quantifies the consumer cost of a single PFD agreement using a structural model of demand and supply. I study the agreement that delayed generic entry into the U.S. market for AndroGel®, the leading testosterone replacement therapy (TRT), from 2006 until 2015. I choose this case for three reasons. First, it is the agreement that reached the Supreme Court in *FTC v. Actavis*. Second, the payments and the stipulated entry dates are documented in the court record. Third, the delay lasted nine years, during which AndroGel® held more than 60% of the market with margins above 65% (AbbVie Inc. v. FTC, 2020). Moreover, generic versions eventually entered in 2015, so the post-entry market is observed and can be used to discipline the counterfactual.

I proceed in three steps. First, I estimate a two-level nested logit demand model on monthly product-level data covering all topical TRT products in the United States from January 2009 to November 2023. Consumers, together with their prescribing physicians, first choose a pharmaceutical form (gel pump, gel packet, patch, or solution) and then a branded or generic version within that form. This ordering follows the institutional sequence of the prescription decision, and it is also supported by the data, since the

nesting parameters are statistically significant and satisfy the restrictions of random utility theory, whereas the reversed hierarchy is rejected. Prices and within-nest shares are instrumented with BLP-style, within-nest, and differentiation instruments (Berry, 1994; Berry et al., 1995; Gandhi and Houde, 2019). Second, I recover marginal costs from the Bertrand-Nash first-order conditions of multi-product firms. Third, I simulate the counterfactual market. For every month before actual generic entry, I add to the choice set the eight generic products that the settling firms eventually launched, assign them the characteristics observed at launch and a marginal cost equal to that of the corresponding AndroGel® product scaled by the generic-to-brand cost ratio observed after actual entry, and solve for the new equilibrium price vector.

The agreement cost consumers $6,451 million between January 2009 and December 2014 ($89.6 million per month), a 43.6% increase in consumer surplus relative to the status quo. With timely entry, generics would have been available at prices 43–50% below the branded product, and the branded price of the gel pump would have fallen by 14–16%. In the gel packet segment the branded price would instead have barely changed (a decrease of 0.4–1.1%). Therefore, the main channel of consumer harm is not only the absence of a branded price cut, but the absence of a cheaper alternative. In other words, the welfare loss comes mainly from the restricted choice set. The losses are concentrated in the two forms in which AndroGel® was marketed, with negligible spillovers to forms not covered by the agreement. Because the data start in 2009 while the delay started in 2006, and because only the products of the settling firms are introduced, the estimate should be interpreted as a lower bound on the true harm.

The supply side of the model shows why the agreement was signed. Under early entry, the branded manufacturer's variable profits over 2009–2014 would have been $2,378 million, against $4,252 million in the status quo. Net of the at most $252 million it paid to the generic challengers ($19–30 million per year for Actavis, $12 million for Paddock and $60 million for Par), the brand retained $1,622 million more than under early entry. For the challengers the comparison is less uniform. Par received more than twice the profits it would have earned by entering, while the foregone profits of Actavis and Paddock exceed their payments, a gap that reflects the 180-day exclusivity of the first filer, which allowed the brand to block all challengers by paying Actavis, and the litigation risk they faced.

This paper makes three contributions. The first one is methodological. The closest paper, Bokhari (2013), quantifies the price effects of a PFD agreement in the ADHD market by removing the settling generics from the observed choice set, so that the delayed-entry world is mimicked by deleting products. I do a complementary exercise and add fictitious generic products to the pre-entry markets. This approach has two advantages. First, it answers directly the policy-relevant question of what would have happened with earlier entry. Second, it lets every incumbent re-optimize its price in response to the new competitors, rather than holding other prices fixed. Moreover, the post-entry period

provides an internal validation, since solving the model in the months after 2015, where the generics are actually observed, reproduces observed prices to numerical precision.

The second contribution is the estimate itself. In the FTC's litigation over AndroGel®, the only monetary figures produced were the brand's profits from delayed generic entry. The FTC's expert estimated $1.2 billion, the district court awarded $448 million, and the Third Circuit reversed the award because the FTC Act does not authorize disgorgement (AbbVie Inc. v. FTC, 2020). No structural estimate of consumer welfare losses was produced. The framework developed here provides one, and it can be replicated for other settlements with product-level sales data and the information contained in the court record.

The third contribution is evidence on where the harm from delayed entry comes from. Reduced-form studies of PFD agreements document that they raise prices and lower quantities (Helland and Seabury, 2016), and theoretical work debates whether they can ever benefit consumers, for instance by avoiding costly litigation or by inducing earlier entry than a court would have granted (Shapiro, 2003; Lemley and Shapiro, 2005; Manganelli, 2021). My results show that in this market the welfare loss operates mainly through the choice set, with branded price reductions confined to the gel pump, that it is confined to the forms covered by the agreement, and that it is about three times larger than the profits the challengers gave up. These quantities are directly relevant for a rule-of-reason assessment.

The paper connects three strands of literature. On PFD agreements, theoretical work has studied entry timing and follow-on products (Lemus and Temnyalov, 2020), the stability of agreements with several potential entrants and authorized generics (Bokhari et al., 2020), and the effect of prohibiting settlements on licensing (Palikot and Pietola, 2023). Empirical work is scarcer. Bokhari (2013) finds price effects of 4–5% in the Adderall case, and Saha and Xu (2023) estimate that delayed entry yields the branded industry between $17.6 and $22.7 billion in revenue. On generic entry, Scott Morton (1999) and Grabowski and Kyle (2007) document what attracts entry, Reiffen and Ward (2005) show that generic prices fall with the number of entrants, and Frank et al. (2021) find that price competition remains the norm in large generic markets. On structural demand for pharmaceuticals, the nested logit has been used to study substitution between branded and generic drugs (Ellison et al., 1997), patent enforcement (Dutta, 2011), reference pricing (Kaiser et al., 2014), parallel trade (Duso et al., 2014; Méndez, 2018), and the welfare effects of generic and me-too entry (Bokhari and Fournier, 2013; Arcidiacono et al., 2013; Dunn, 2012). I bring these tools to the antitrust evaluation of a specific settlement. As a robustness check, I also estimate a random coefficients logit; its counterfactual predictions are qualitatively consistent, but the random coefficients are weakly identified in a single national market observed over time, which is why the nested logit, whose nesting parameters capture the relevant heterogeneity (Grigolon and Verboven, 2014), is

the main specification.

The rest of the paper is organized as follows. Section 2 describes the U.S. market for testosterone replacement therapy and Section 3 the AndroGel® settlement and its litigation. Section 4 presents the data. Section 5 sets out the demand and supply model and the counterfactual algorithm. Section 6 reports demand estimates, substitution patterns, and the welfare and profit effects of delayed entry. Section 7 concludes.

2 Industry background: The pharmaceutical market of Testosterone in the US

Hypogonadism is a condition characterized by low levels of the male hormone testosterone. Men who suffer from this condition do not produce enough testosterone, that plays a key role in masculine growth and sperm production. There are two types of hypogonadism: the primary (also known as primary testicular failure), which originates from a problem in the testicles, and the secondary, which indicates a problem in the hypothalamus or the pituitary gland. Either type of hypogonadism can be caused by an inherited trait or it can happen later in life, acquiring it through an injury or an infection. Hypogonadism can develop in serious issues such as undescended testicles, pituitary disorders, and inflammatory diseases, among others (Mayo Clinic, 2025).

This disease is usually treated with TRT which is usually dispensed in many forms such as injectables, gels and patches, among others. The two most popular options are transdermal gel patches and injectable vials through intramuscular injections. The patient needs a prescription from the doctor to obtain this drug, and usually a maximum amount of 6 months is prescribed (Sizar et al., 2024). Hypogonadism is a condition that has an estimated prevalence of approximately the 38% of men in the United States over 45 years old (Mulligan et al., 2006).

The TRT market in the United States comprises several pharmaceutical forms of administration, such as transdermal products, injectable vials, nasal and oral products among others. These forms differ in their route of administration, dosing frequency, and clinical profile. The choice of form is primarily a clinical decision made by the prescribing physician based on patient-specific factors such as absorption characteristics, needle aversion, and comorbidities (Bhasin et al., 2018). Once the form is determined, the choice between branded and generic versions, when available, is largely driven by cost considerations, insurance formulary placement, and pharmacy substitution rules.

The main prescribed form in the United States is the vials injectable segment, followed by the topical forms (Bhat and Dobs, 2022). The PFD agreement at the center of this paper affected only the topical gel forms in which AndroGel® was marketed: the topical gel pump and the transdermal gel packet forms. The remaining forms were not subject

to the agreement and had their own competitive dynamics. Nevertheless, because all forms treat the same underlying condition and compete within the same therapeutic class, competitive effects in the PFD-affected segments can propagate to other forms through cross-form substitution. The demand model accounts for this through the nested logit structure, where the upper nest captures substitution across forms.

The case of generic entry for AndroGel® raised many legal concerns. The FTC sued Actavis and other generic firms for preventing early generic entry into the testosterone market by means of a PFD agreement before patent expiration through a Paragraph IV challenge (Federal Trade Commission, 2013).

3 The case for AndroGel®

In 1995 Solvay Pharmaceuticals entered into a licensing agreement with Besins to market AndroGel® in the U.S. (Dickey et al., 2010). In 1999 Solvay filed a New Drug Application (NDA) for the brand-name drug AndroGel®[1]. The FDA granted marketing approval for the NDA #021015 in the 2000 and began being marketed that same year. Because of this drug being the first testosterone gel approved in the US, it quickly became a blockbuster, attracting generic entry.

In 2003 Actavis (then known as Watson Pharmaceuticals) and Paddock Laboratories both filed an ANDA certifying that under the Paragraph IV of the Hatch-Waxman act, their generic version of the drug did not infringe the AndroGel® patent. Par Pharmaceutical, a third potential generic entrant, did not file a Paragraph IV challenge, but it agreed to share litigation costs with Paddock (Federal Trade Commission, 2012).

Solvay sued Actavis and Paddock for patent infringement within the 45-day window allowed by the Hatch-Waxman Act, which triggered an automatic 30-month stay of FDA approval. During two years, Solvay, Actavis and Paddock litigated regarding the #894 AndroGel® patent. In January 2006 the FDA approved Actavis's ANDA, and Solvay was threatened with losing its monopoly over AndroGel® because of the imminent generic entry into the market of these firms (Schipper, 2013).

Therefore, Solvay offered generics to settle out of court. Since Actavis was the first-to-file, the settlement stipulated that Actavis would not bring its generic version of AndroGel® to the market until 31st of August, 2015, implying a 9-year delay in the expected entry of generics, but 65 months before Solvay's patent expired. This agreement would hold until the date of the settlement, unless someone else marketed a generic sooner. Note that, Actavis's stipulated entry date is 180 days earlier than the date of entry of the second entrant, Par Pharmaceuticals, who was scheduled to enter on February 26,2016 (Hemphill, 2009). In exchange, Actavis would receive $19–30 million per year, Paddock would receive

[1]In Appendix C, I provide further explanation for the mechanisms of generic entry in the United States

$12 million in total, and Par would receive $60 million in total. The companies argued that these payments were a compensation for other services that the generics promised to perform, however, according to the FTC, these payments were made to compensate generics not to compete against AndroGel® until 2015 (Federal Trade Commission, 2013).

On January 2009 the Federal Trade Commission (FTC) filed a lawsuit against all these firms by unlawfully agreeing to

"*Share in Solvay's monopoly profits, abandon their patent challenges, and refrain from launching their low-cost generic products to compete with AndroGel® for nine years*" (Federal Trade Commission, 2013).

The case went through two lower courts (District Court for the Northern District of Georgia and United States Court of Appeals for the Eleventh Circuit) that argued that these payments were lawful and they did not restrain competition at all. However, in 2013, the case arrived to the Supreme Court of the United States. This court ruled that these agreements are subject to antitrust scrutiny and should be assessed with a "rule of reason". This opened a benchmark for assessing these cases one by one, instead of treating all of them as anticompetitive "per se".

Generic entry into the TRT market finally occurred in 2015, when Perrigo, which had acquired Paddock Laboratories in 2011, launched its generic version of AndroGel® on January 2015, in both the transdermal gel packet and topical gel pump forms. Par Pharmaceuticals followed with its own version of the generic transdermal gel packet versions, in September 2015, and Actavis launched its own versions of transdermal and topical gel pump products in November 2015[2]. Further generic entry occurred in 2018, when other firms such as Padagis and Zydus Pharma (among others) entered the market.

The case was remanded to the district court and the trial was scheduled to begin on March 2019. In 2019, the FTC reached a settlement agreement with Abbvie, the current owner of Solvay, to resolve the outstanding litigation. Under this settlement, Abbvie is prohibited from entering into certain anti-competitive agreements with other firms such as: (i) making side deals with other firms where the generic receives a compensation for settling patent litigation and (ii) no Authorized-Generic agreement where a brand company agrees not to compete with an authorized version of a generic drug (Federal Trade Commission, 2019).

3.1 The structure of the market

For the purposes of this paper, I define the relevant product market as the set of topical testosterone replacement therapies , comprising four pharmaceutical forms: topical gel pump, transdermal gel packet, patch/disc, and topical solution pump. I exclude injectable

[2]These observed entry dates in my dataset differ slightly from the dates stipulated in the settlement agreement, as the actual timing of commercial launch depends on manufacturing readiness, FDA final approval, and individual firm decisions.

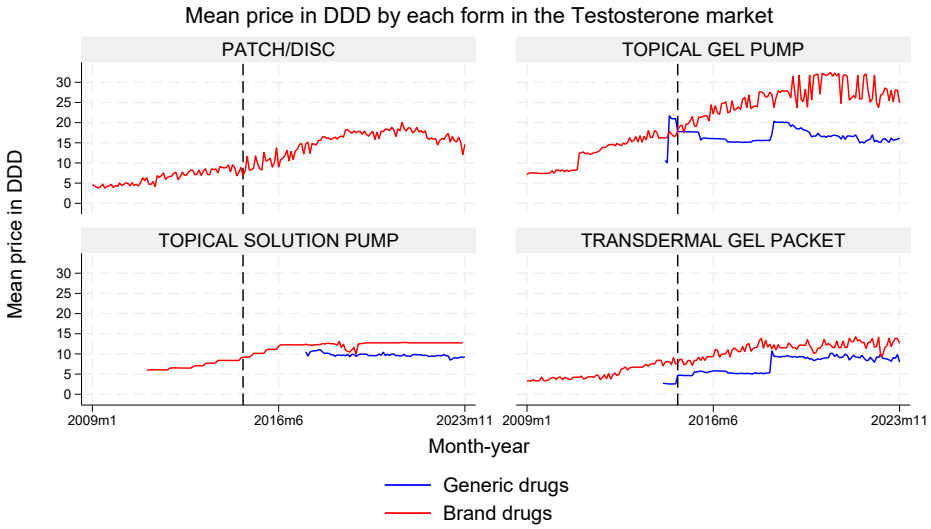


Figure 1: Price evolution by form. The black dashed line indicates first entry of generics in the market by Perrigo Pharmaceuticals.

vials from the primary analysis for two reasons. First, injectable formulations differ fundamentally from topical products in their route of administration, dosing frequency, and clinical profile, and clinical practice guidelines treat them as a distinct therapeutic modality[3]. Second, and more importantly, the firms involved in the PFD agreement themselves did not consider injectables as relevant competitors to AndroGel®. During trial, AbbVie's Director of Marketing testified that AbbVie did not consider injectables to be competition, and identified other products such as Axiron, Fortesta, and Testim as AndroGel®'s direct competitors (AbbVie Inc. v. FTC, 2020).[4]

This market definition is further supported by the trial record. The district court found that AbbVie itself tracked the TRT market as the relevant competitive arena: AbbVie reported AndroGel®'s sales within the TRT market to its Board of Directors and to investors, and multiple internal documents introduced into evidence demonstrate that AbbVie monitored other TRTs as direct competitors[5]. AndroGel® was the dominant product in this market, holding a share above 60% from 2011 through the end of 2014, with no other branded TRT ever reaching 10% during this period[6]. AbbVie maintained profit margins above 65% even while offering substantial rebates. Although three brand-name TRT products entered the market between 2011 and 2014, Fortesta (Endo Pharmaceuticals, February 2011), Axiron (Eli Lilly, March 2011), and Vogelxo (Upsher-Smith, July 2014),

[3]As the district court concluded, "even assuming reasonable interchangeability, there is little cross-elasticity of demand between AndroGel and injectables to include injectables in the relevant market." *FTC v. AbbVie Inc.*, No. 14-cv-05151, Findings of Fact and Conclusions of Law page 68 (E.D. Pa. June 29, 2018).

[4]Appendix to Petition for Certiorari, *AbbVie Inc. v. FTC*, 2020, pp. 73a.

[5]Appendix to Petition for Certiorari, *AbbVie Inc. v. FTC*, 2020, pp. 147a.

[6]Appendix to Petition for Certiorari, *AbbVie Inc. v. FTC*, 2020, pp. 74a.

the court found that these products did not pose significant competitive pressure, as each maintained a relatively small market share compared to AndroGel®[7]. The court concluded that significant barriers to entry, including capital, technical, regulatory, and legal hurdles, protected AbbVie's dominant position in the TRT segment (AbbVie Inc. v. FTC, 2020).

This evidence supports the modeling choice of treating topical forms as the relevant market and structuring the demand model around substitution within and across these forms, while excluding injectables from the main specification.

The structure of the TRT market, shown in Figure 1 reveals a pattern that could be explained by the so-called "Generic Competition Paradox" (Vandoros and Kanavos, 2013). Branded prices rise steadily before generic entry, however, after generic competition happens, the branded price increases sharply. This phenomenon might be due to brand loyalty, where branded manufacturers raise their prices in response to generic entry, in order to capture the inelastic portion of customers and differentiate from the generic alternatives (Frank and Salkever, 1997). Patients who are accustomed to using the branded TRT product may be reluctant to switch to a generic version, owing to uncertainty about the efficacy and safety of the generic product. Moreover, physicians might still continue to prescribe the branded versions even if a generic alternative is available, driven by this brand loyalty. Patients' and physicians' preferences for the established branded product, despite the availability of lower-cost generic options, can create a situation where the branded product's price rises after generic entry, and individuals keep demanding and consuming these branded products, rather than switching to generic and cheaper versions of the product, as it would be expected.

The United States market for hypogonadism drugs is large. In 2002, the pharmaceutical sales for testosterone were $324 million, and it grew up to $2 billion in 2012 (Busnelli et al., 2017). According to the Global Strategic Business Report - TRT, the market for testosterone is projected to reach $2.5 billion by 2030 (Research and Markets, 2025).

Most private insurance plans cover TRT if it is deemed medically necessary (for example, for a confirmed diagnosis of hypogonadism). However, many insurers will not cover TRT for age-related hormone decline or other non-essential uses. Insurers usually require a specific criteria to be met, such as blood tests showing low testosterone levels and a doctor's documentation, before approving coverage. Medicare will cover testosterone therapy in certain situations, but the coverage depends on how and where the drug is provided. Medicaid also provides coverage for testosterone prescriptions when they are medically necessary.

Figure 2 displays the evolution of market shares by pharmaceutical form in the TRT market from January 2009 to November 2023. At the beginning of the period, Transdermal Gel Packets dominated the market with a share close to 65%, while Topical Gel Pumps accounted for approximately 30%. Over time, a clear reallocation in market structure

[7]Appendix to Petition for Certiorari, *AbbVie Inc. v. FTC*, 2020, pp. 154a-155a.

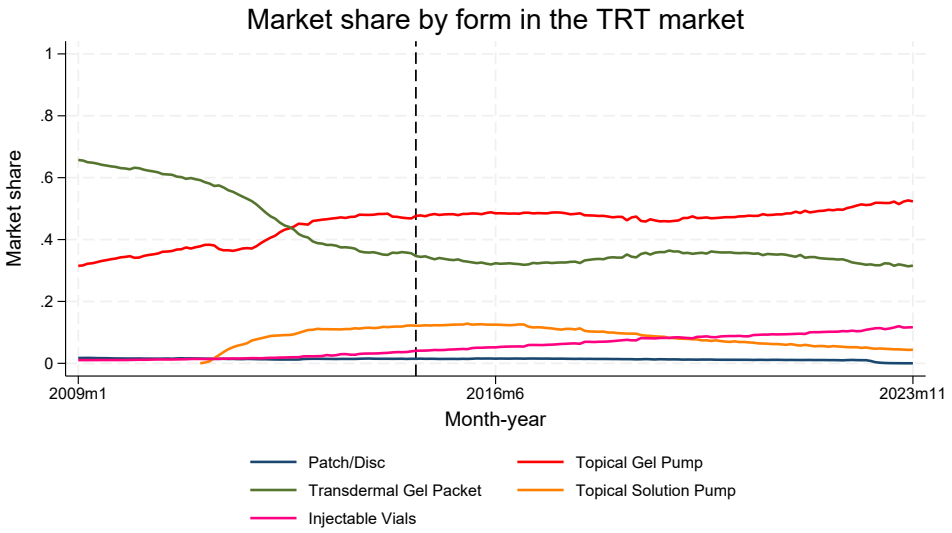


Figure 2: Share evolution by form and month. The black dashed line indicates first entry of generics in the market by Perrigo Pharmaceuticals. Other non-relevant forms for the market are not shown due to negligible shares.

emerges. The share of gel packets declines steadily, stabilizing at around 30% in the later years, whereas gel pumps experience a persistent increase and become the dominant form by the end of the sample, reaching roughly 50% of the market.

The remaining formulations maintain substantially smaller shares. Topical Solution Pumps exhibit a temporary increase followed by a gradual decline. Injectable Vials grow steadily throughout the period but remain below 15% of total market share. Patch/Disc products lose relevance continuously and become nearly negligible by the end of the sample.

4 Data and descriptive statistics

The dataset contains information about the Testosterone molecule, and it spans from January 2009 until November 2023.

Data used in this paper is taken from Bloomberg Symphony Health. It provides weekly information on sales and volumes at the product level for several products within the Testosterone molecule. For each product, I observe the total number of units dispensed, and the acquisition cost reported by pharmacies, which reflects the amount paid to wholesalers. I compute unit prices and harmonize products by transforming prices using DDD equivalents. I aggregate the dataset to monthly level.

A limitation of the data is that prices reflect acquisition costs by pharmacies (i.e. price paid to wholesalers), rather than final consumer prices. Since pharmacies typically apply a markup when dispensing drugs, observed prices will likely be lower than actual prices faced by patients. Moreover, I am unable to retrieve data before 1, Jan 2009, but the

Table 1: Dataset observations by drug form and drug indicator (Generic or Brand)

Form	Generic	Branded	Total
Topical gel pump	958	677	1,635
Transdermal gel packet	1,208	1,121	2,329
Patch/Disc	0	860	860
Topical solution pump	366	112	478
Total	2,532	2,770	5,302

delay period effectively started in 2006. Therefore, the estimated price levels should be interpreted as conservative, representing a lower bound.

I also observe drug characteristics such as name, strength, form, package size, manufacturer and branding status (brand or generic drug). A product in the dataset is defined as a combination of these attributes.

Overall, the dataset contains 30 firms that produce a total of 79 products, 36 branded products and 43 generic products. These products are presented in the four main forms of the TRT transdermal market: topical gel pump, a gel formulation dispensed through a pump bottle for application on the skin,transdermal gel packet, gel packaged in single-use packets, also applied topically to deliver the drug through the skin, patches, adhesive patches or discs applied to the skin, releasing the drug gradually over time, and topical solution pump, a liquid solution delivered via a pump for skin application, similar in use to topical gels but with a different base. The number of products in each market varies month by month, due to product entry and exit into the market[8].

Table 2 reports the descriptive statistics for the most important variables used in the model, including the prices in different units, shares and within-market shares. Prices vary considerably across forms. In the lowest part of the table I report both the number of products and the number of firms producing those products, which are used to construct the instrumental variables for prices and within-market shares.

5 Empirical strategy

5.1 Demand model

In this section, I first briefly describe my demand and supply-side equations, and then focus on the counterfactual. This demand specification is based in previous work from Berry (1994), Verboven (1996) and Duso et al. (2014).

[8]Further descriptive evidence of the market can be found in Appendix D.

Table 2: Summary statistics, hypogonadism drugs (2009 - 2023)

	Total		Branded		Generics	
	Mean	SD	Mean	SD	Mean	SD
s_{jt}	0.003	0.009	0.004	0.012	0.001	0.002
$s_{j\|h_{g,t}}$	0.18	0.28	0.23	0.35	0.12	0.16
$s_{h\|g,t}$	0.78	0.33	0.73	0.38	0.83	0.25
Price/product (DDD/USD)						
Total (p_{jt})	12.09	8.79	12.59	8.88	11.53	8.65
Topical gel pump	18.04	10.33	19.97	10.69	16.68	9.84
Transdermal gel packet	8.43	6.94	8.88	7.23	8.02	6.63
Patch/Disc	11.99	5.77	11.99	5.77	–	–
Topical solution pump	9.70	2.44	9.84	2.50	9.66	2.42
# of products	79		36		43	
# of products in Topical gel pump	23		8		15	
# of products in Transdermal gel packet	30		10		20	
# of products in Patch/Disc	17		17		0	
# of products in Topical solution pump	9		1		8	
# of firms	30		12		20	
# of firms in Topical gel pump	17		6		13	
# of firms in Transdermal gel packet	14		5		11	
# of firms in Patch/Disc	5		5		0	
# of firms in Topical solution pump	9		1		8	

I report the descriptive statistics for the 79 drugs treating hypogonadism in different forms in the United States, over 14 years and 11 months. I use the nest g to define the drug form level, and the nest h to define whether the drug is branded or generic. s_{jt} is the overall market share of product j in market t, $s_{j|h_{g,t}}$ is the market share of the product within the inner nest (indicator), and $s_{h|g,t}$ is the market share of the inner nest (indicator) within the outer nest (drug form). All values are based on my own calculations with data from *Bloomberg Symphony Health*; 5,302 observations. DDD: defined daily doses.

5.1.1 Joint utility maximization

In each market $t = 1, 2, ..., 179$, the consumer chooses one of the $j = 1, 2, ..., J_t$ products. Therefore, in my dataset, I consider each observation period as an independent market, reflecting the set of available products and consumer choices in that specific month. The consumer might choose to buy the outside option, which is either not buying any drug at all and/or treat hypogonadism with lifestyle changes. The consumer[9] i's conditional indirect utility function for product j in market/month t is:

$$u_{ijt} = \alpha \cdot p_{jt} + \beta \cdot x_j + \xi_j + \epsilon_{ijt} \tag{1}$$

where p_{jt} is the price of the product j in market t expressed in \$/Defined Daily Dose (DDD)[10], and x_j are the product characteristics, such as drug strength and drug package size, both expressed in terms of DDD. ξ_j contains all the characteristics that are observed by firms and consumers but not by the econometrician, and ϵ_{ijt} is a random shock to utility, distributed as i.i.d., and I assume that it follows the distributional assumptions of the two-level nested logit model.

The expected utility parameter is:

$$\delta_{jt} = \alpha \cdot p_{jt} + \beta \cdot x_j + \xi_j \tag{2}$$

5.1.2 Market size and outside good

Low testosterone prevalence in the US is estimated to be in a range of about 30% - 40% of male population, according to several studies (Liu et al., 2022), however, hypogonadism prevalence is estimated $\sim$ 10% in men aged 50-59, and $\sim$ 20% in men 60-69 (Harman et al., 2001). Some sources suggest that hypogonadism affects 4 to 5 million men in the US, even though it may occur in men at any age (Boston University School of Medicine, 2025). According to the 2020 U.S. Census "Sex by Age" B01001 there are about 75.7 million men aged 35-79. Assuming a 10% prevalence, this turns to the 7.5 potential population.

Potential market size M_t is defined per each market using a weighted sum of the forms observed in that market in the dataset. For each market, I compute the relative market share of each drug form by calculating the share of products of each form in the total product quantity sold. Then, each form is assigned a DDD weight based on the WHO guidelines for testosterone daily intake. I then compute a weighted average of DDDs across forms, multiply it by the potential population, and define this as the effective potential

[9] While physicians are likely to influence prescribing decisions for testosterone therapies, I abstract from modeling their preferences explicitly and instead assume that demand reflects the joint outcome of patient and prescriber choices.

[10] The Defined Daily Dose is the assumed average maintenance dose per day for a drug used for its main indication in adults, as defined by the World Health Organization (WHO). It is a standardized unit that allows comparison of drug consumption across different products, strengths, and dosages.

market size M_t. Therefore, the total potential US market in my model is

$$M_t = \sum_f (\text{DDD}_f \times \text{Share}_{ft}) \times 7{,}500{,}000 \tag{3}$$

where DDD_f is the Defined Daily Dose assigned to form f, and Share_{ft} is the proportion of products in form f in market t. This method accounts for heterogeneity in form intensity and product composition over time[11].

5.1.3 Nesting structure

In the market for testosterone drugs the consumer/physician must choose which form of the drug is more suitable for him, based on the patients' physical condition, individual preferences, side-effects, age, and medical history. Then he chooses to buy the branded or generic version, depending on the availability. The discrete-choice framework underlying this model assumes that each patient uses a single TRT product at a time, which is consistent with clinical practice guidelines that recommend selecting a single testosterone formulation based on patient-specific factors and do not contemplate concurrent use of multiple formulations (Bhasin et al., 2018). This nesting hierarchy (form first, then branded versus generic status), is motivated by the institutional features of the TRT market. The choice of pharmaceutical form is primarily a clinical decision from the physician. They choose the delivery method based on patient-specific factors such as patient preference, treatment burden or absorption characteristics (Bhasin et al., 2018). Once the form and dosage is determined, the choice between branded and generic versions is largely a cost-driven decision, often influenced by insurance formularies and pharmacy substitution rules. This ordering reflects the sequential nature of prescribing: clinical suitability and preferences of the patient determine the form, and economic considerations determine the brand/generic choice within that form. Therefore, I exploit this natural ordering of choice in my nesting structure.

The first level of G nests (upper nest) $g = 1, 2, 3, 4$ comprises the form of the drug, while the second level of H_g nests (lower nest) $H_g, h = 1, 2$ comprises whether the drug, within that particular form, is branded or generic[12]. Because of the reasons mentioned in Section 3, injectable vials form is not included in the final sample since it is a form

[11]The demand estimates are robust to alternative market size definitions. Using a prevalence rate of 5% (potential population of 3.79 million) and 15% (11.35 million) instead of the baseline 10% (7.5 million), the estimated parameters remain virtually unchanged.

[12]As a specification check, I also estimated the two-level nested logit with the reversed nesting hierarchy, placing the branded/generic distinction at the upper level and pharmaceutical form at the lower level. Under this alternative structure, the upper-nest parameter is not statistically significant, indicating that the branded/generic grouping does not capture meaningful preference correlation at the top level of the choice hierarchy. This supports the baseline specification in which form constitutes the upper nest, consistent with the institutional ordering whereby the clinical choice of pharmaceutical form precedes the economic choice between branded and generic alternatives.

not considered to be competing directly with AndroGel® and all transdermal products included in the settlement. Following Verboven (1996) the estimation equation is:

$$ln(s_{jt}) - ln(s_{0t}) = \alpha p_{jt} + \beta \mathbf{x}_j + \sigma_1 ln(s_{j|hg,t}) + \sigma_2 ln(s_{h|g,t}) + \xi_{jt} \tag{4}$$

where $s_{jt} = q_{jt}/M_t$ and $s_{0t} = 1 - \sum_{j=1}^{J_t}(q_{jt}/M_t)$ are the market shares of drug j in market t and the market shares of the outside good 0 in market t, q_{jt} are sales in DDD and p_{jt} is the price in DDD. The inner-group market shares are defined as follows:

$$s_{j|hg,t} = \frac{q_{jt}}{\sum_{j \in H_g} q_{jt}} \tag{5}$$

$$s_{h|g,t} = \frac{\sum_{j \in H_g} q_{jt}}{\sum_{g=1}^{G} \sum_{j \in H_g} q_{jt}} \tag{6}$$

The estimable equations of the one level nested logit and the plain logit are the same, but for the one level nested logit I omit the within market share term $s_{j|hg,t}$ and for the plain logit I omit both $s_{j|hg,t}$ and $s_{h|g,t}$.

One of the main reasons for going structural in this setting is the appropriateness for the counterfactual exercise. A reduced-form approach such as differences-in-differences or an event study around 2015 could estimate the effect of generic entry at the time it actually occurred, but it cannot answer the policy-relevant counterfactual: what would equilibrium prices and consumer welfare have been had entry occurred years earlier, in a different competitive environment? This requires changing the choice set available to consumers, solving for the endogenous price responses of all incumbent firms under the new market structure, and computing consumer surplus in monetary terms, none of which a reduced-form design can deliver without observing the counterfactual directly. Moreover, the absence of a plausible control group, since the PFD agreement affected all topical TRT forms nationally and no comparable therapeutic market was subject to a similar delay during this period, makes reduced-form identification particularly difficult.

The structural approach replaces these requirements with a different set of assumptions. The counterfactual simulation assumes that consumer preferences (the demand parameters $\hat{\alpha}$, $\hat{\sigma}_1$, $\hat{\sigma}_2$, and the product-level unobservables ξ_{jt}) and marginal costs remain stable across the delay period, so that the estimated demand and supply primitives can be applied to the augmented choice set. The key identifying advantage of the structural model is that, conditional on these primitives, it endogenously determines counterfactual equilibrium prices through the Bertrand-Nash first-order conditions, rather than assuming that observed post-entry price responses would have been identical under earlier entry.

A potential concern of using this particular demand system is that the nested logit imposes substitution patterns that are governed by the nesting structure. Since the counterfactual exercise in this paper involves introducing generic products into specific

forms, the welfare effects could partly reflect the imposed nesting rather than underlying demand behavior. To address this concern, I also estimate a Random Coefficients Logit (RCL) model, and perform the same counterfactual as a robustness check (reported in Appendices E and F). The RCL allows for unrestricted substitution patterns across all products, as heterogeneity in consumer preferences over product characteristics generates flexible cross-price elasticities without imposing a pre-specified grouping structure (Berry et al., 1995).

However, the RCL faces a fundamental identification challenge in this setting. The standard source of identification for the random coefficients in the model is the cross-market variation in product characteristics and prices. Essentially, observing how consumers in different markets respond to different choice sets (Nevo, 2000). However, in this paper, the TRT is a single national market observed over time t, with no geographic variation. As a result, this makes very difficult to identify the random coefficients only from temporal variation in the dataset. Temporal variables such as product entry, exit, and pricing provide very limited information for separating preference heterogeneity from unobserved demand shocks. On the other hand, Grigolon and Verboven (2014) show that nesting parameters capture an important source of consumer heterogeneity, and that the nested logit yields robust counterfactual predictions to obtain reliable policy conclusions. Therefore, I will consider the 2 level nested logit as the main specification for this paper.

5.1.4 Instruments and identification

In this setting, price and the inner market shares are endogenous. This is because the term ξ_j represents the product characteristics observed by the consumer and the firms, but not by the econometrician. These product characteristics are taken into account by firms when they set their prices, which implies that prices and inner-group market shares will be correlated with the error term, making the price and the inner shares endogenous. Therefore, I use several instruments in order to alleviate this problem.

The first set of instruments consists on the BLP style instruments (Berry (1994) and Berry et al. (1995)). The first one is the sum of package size across rival firms' products in the market, and the second one is the sum of the drug strength across own-firm products in the market, excluding its own product.

The second set of instruments consists of within-nest instruments, following Verboven (1996). Specifically, I compute the mean drug strength of rival firms' products within the same pharmaceutical form in that particular market, which is set to zero when product j has no rivals in its form.

The third set of instruments consists of differentiation instruments following Gandhi and Houde (2019). For each product j in market t, I count the number of rival and own-firm products that are located within one standard deviation of product j along each

characteristic dimension x_j (drug strength and drug package size), or within a unit distance of product j in the joint space of both standardized characteristics. Specifically, I use three local differentiation instruments: the number of rival products within one standard deviation in drug strength, the number of own-firm products within one standard deviation in drug strength, and the number of own-firm products located within a Euclidean distance of one from product j in the space of standardized drug strength and package size, that is, own-firm products k with $\sqrt{(\tilde{x}_k^{str} - \tilde{x}_j^{str})^2 + (\tilde{x}_k^{size} - \tilde{x}_j^{size})^2} < 1$, where each characteristic $\tilde{x}$ is standardized by its sample mean and standard deviation. This last instrument measures both drug size and drug strength caractheristics jointly.

The main assumption is that these instruments are correlated with the inner market shares and prices, but are uncorrelated with the product-specific error term ξ_j[13].

5.1.5 Elasticities

In the computation of the elasticities I closely follow Berry (1994), Verboven (1996) and Duso et al. (2014) for calculating:

- The own-price elasticity
- The cross-price elasticity within the same subnest but different nest
- The cross-price elasticity within the same nest
- The cross-price elasticities outside the nest

For each market t I compute one matrix of price elasticities, resulting in 179 matrices of elasticities, each of one of size $(J_t \times J_t)$. Each matrix has a different size because of continuous entry and exit of products in the market. The formulas for computing these matrices can be found in the appendix.

The nested logit model is restrictive when representing substitution outside or within groups, however it is quite flexible when it comes to asymmetry of cross-price elasticities, because these only depend on the estimates structural parameters and the price and market shares of the group.

5.2 Consumer Surplus

Following the well-known formula of the consumer surplus in the Logit model (Train, 2009), as well as the formulas for the one and two-level nested logit model (Dutta 2011; Duso et al. 2014) , I am able to compute the Consumer Surplus (CS) as:

[13] A discussion of the validity of the exclusion restrictions, as well as the first-stage regression results, is provided in Appendix A.

$$CS_t(p_t) = \frac{1}{\hat{\alpha}} M_t \ln\left(1 + \sum_{g=1}^{G}\left(\sum_{h=1}^{H_g} D_{h|g,t}^{\frac{(1-\sigma_1)}{(1-\sigma_2)}}\right)^{(1-\sigma_2)}\right) \tag{7}$$

where

$$D_{h|g,t} = \sum_{j \in h_g} \exp\left(\frac{\hat{\delta}_{jt}}{1-\sigma_1}\right)$$

and $\hat{\delta_{jt}}$ are the fitted values of equation (2) using the estimated demand parameters.

5.3 Supply side

5.3.1 Multi-product firm supply side

The profit function of the multi-product firm $f(f = 1, ..., F)$ active in time/market t that manufactures a subset $\mathcal{F}_{ft}$ of the J products is:

$$\Pi_{ft} = \sum_{j \in \mathcal{F}_{ft}} (p_{jt} - c_{jt}) s_{jt} \cdot M_t - \sum_{j \in \mathcal{F}_{ft}} C_{jt} \tag{8}$$

where s_{jt} is the share of product j in time/market t, p_{jt} is the price of the product (in DDD) and c_{jt} is the marginal cost of that firm. C_{jt} is the fixed cost of the product.

5.3.2 First Order Conditions

Each firm maximizes its own profit in each time t, therefore, taking the FOC from equation (8) with respect to p_{jt} we have:

$$s_{jt} + \sum_{k \in \mathcal{F}_{ft}} (p_{kt} - c_{kt}) \cdot \frac{\partial s_{kt}}{\partial p_{jt}} = 0 \tag{9}$$

Following Berry et al. (1995), I can denote the above FOC in vector notation. Let Ω_t^F be the firms' product ownership matrix with structure $(J_t \times J_t)$ with elements $\Omega_t^F(j,k)$ equal to 1 if product j and product k are produced by the same firm in time/market t, and 0 otherwise. Let $\Delta_t = \frac{\partial s_{kt}}{\partial p_{jt}} \odot \Omega_t^F(j,k)$ be a J_t by J_t Jacobian matrix where j indexes rows and k indexes columns. Therefore, equation (9) can be re-written in vector form as:

$$\mathbf{s_t} - \Delta_t(\mathbf{p_t} - \mathbf{c_t}) = 0 \tag{10}$$

where $\mathbf{s_t}$ is the $J_t \times 1$ vector of shares, $\mathbf{p_t}$ is the $J_t \times 1$ vector of prices, and $\mathbf{c_t}$ is the $J_t \times 1$ vector of marginal costs.

Using this, I can back out marginal costs per period as:

$$\mathbf{c_t} = \mathbf{p_t} - \Delta_t^{-1}\mathbf{s_t} \tag{11}$$

where $\Delta_t^{-1}\mathbf{s_t}$ is the predicted markup.

5.3.3 Variable profits

To evaluate the consequences of delayed generic entry from the supply side, I compute variable profits for each firm in each market:

$$VP_{ft} = \sum_{k \in \mathcal{F}_{ft}} (p_{jt} - c_{jt}) \cdot s_{jt} \cdot M_t \tag{12}$$

These profits are aggregated across markets over time and reported in $ million.

5.4 Counterfactual estimation

To quantify PFD damages I answer the counterfactual question: what would market outcomes have been if entry had occurred earlier? To do so, I perform a fixed-point algorithm using the optimality condition on equation (10), artificially introducing eight generic products in the market. These are the products shown in Table 3.

Table 3: Summary of product entries by firm

Firm	Strength*	Package Size*	Form	Observed entry date
Perrigo	12.5/1.25g	75	topical gel pump	2015m1
Perrigo	25mg(1%)	2.5	transdermal gel packet	2015m1
Perrigo	50mg(1%)	5	transdermal gel packet	2015m1
Par Pharma	25mg(1%)	2.5	transdermal gel packet	2015m9
Par Pharma	50mg(1%)	5	transdermal gel packet	2015m9
Actavis Pharma	25mg(1%)	2.5	transdermal gel packet	2015m11
Actavis Pharma	50mg(1%)	5	transdermal gel packet	2015m11
Actavis Pharma	12.5/1.25g	75	topical gel pump	2015m12

*Strength and package size refer to the actual pharmaceutical presentations as marketed, rather than DDD.

The reasons to include these products is that they are the first generics to enter in the market in the status quo, and all of these firms were involved in the PFD agreement.[14] For each market t prior to January 2015, I:

1. Augment the product set in market t by introducing the eight generic products listed in Table 3. These synthetic generics have the same product characteristics (e.g., strength, form) as observed at launch, and are associated with their respective firms.

2. Solve for counterfactual equilibrium prices using a fixed-point iteration on the Bertrand-Nash first-order conditions. For each market t, this involves:

[14] In Appendix G, I provide a robustness check for this algorithm, by testing it against the actual observed entry that happened in 2015.

- Computing the mean utility $\hat{\delta}_{jt}$ of each product using the estimated demand parameters from the NL2 IV model[15], observed characteristics, and current prices;
- Deriving predicted market shares using the two-level nested logit structure (equations 25 to 31);
- Computing product-level markups from the first-order conditions in equation (9), accounting for multi-product firm ownership;
- Updating prices as $p_j^{(n+1)} = c_j + \text{markup}_j^{(n)}$ with damping, and iterating until the price vector converges.[16]

Marginal costs for the existing products are recovered from the observed status quo, using equation (11). However, for the artificially introdced generic products, no cost can be recovered before entry, since they are only observed post entry. Therefore, I compute their marginal cost in month t equal to the marginal cost of the AndroGel® product with the same form, strength and package size in that month, multiplied by the generic-to-brand cost ratio observed after actual entry, defined as the median over 2017–2023 of the ratio between the recovered marginal cost of the corresponding real generic and that of AndroGel®. These ratios range from 0.26 to 0.69 depending on the presentation (0.47 on average), so the fictitious generics enter with the cost advantage over the originator that their real counterparts display once in the market[17].

To create a proper measure of the damage suffered by consumers from the PFD agreement, I define damages as the difference between the consumer surplus with early generic entry (counterfactual) and the consumer surplus under the status quo, all summed during the delay period. More formally the damage suffered by the delay in each period t is:

$$\text{PFD Damages}_t = CS_t^{\text{counterfactual}} - CS_t^{\text{status quo}} \tag{13}$$

The total PFD damages over the delay period are given by the sum over all relevant months t in the delay period:

$$\text{Total PFD Damages} = \sum_{t \in \text{delay period}} \left(CS_t^{\text{counterfactual}} - CS_t^{\text{status quo}} \right) \tag{14}$$

[15] In Appendix F I provide results under the same counterfactual procedure using the parameters from the RC Logit (OI) model.

[16] The algorithm is described in full detail in Appendix B.

[17] Using mean post-entry levels is another common approach in the literature. For instance Di Giacomo (2008) applies it in the yogurt industry, and Hausman and Leonard (2002) in the tissue industry, which are quite homogeneous markets in terms of costs. Following this approach in a pharmaceutical market, would mean that generics would enter at a price directly above the brand level before 2015, since recovered marginal costs are two to five times higher after 2015 than before. Moreover, there is a related body of work showing that generic drugs have a cost advantage with respect to branded drugs (Arcidiacono et al., 2013; Dubois and Lasio, 2018).

6 Results

6.1 Demand estimates

Table 4: Demand estimation results

Variable	NL1	NL1 IV	NL2	NL2 IV	RC Logit	RC Logit (OI)
Intercept	-2.07*** (0.07)	-0.83** (0.33)	-2.10*** (0.07)	-0.77** (0.31)	–	–
Size (in DDD)	-0.03*** (0.00)	-0.02*** (0.00)	-0.03*** (0.00)	-0.02*** (0.00)	0.07*** (0.01)	0.08*** (0.01)
Strength (in DDD)	-1.05*** (0.06)	-2.67*** (0.13)	-1.03*** (0.06)	-2.65*** (0.12)	-6.17*** (0.61)	-6.66*** (0.42)
σ_1	–	–	0.99*** (0.00)	0.85*** (0.03)	–	–
σ_2	0.99*** (0.00)	0.82*** (0.03)	0.86*** (0.01)	0.61*** (0.10)	–	–
α	-0.06*** (0.00)	-0.18*** (0.01)	-0.06*** (0.00)	-0.18*** (0.01)	-0.60*** (0.09)	-0.85*** (0.11)
σ_α	–	–	–	–	0.00 (4.37)	0.15*** (0.04)
σ_B	–	–	–	–	0.00 (64.31)	0.00
σ_G	–	–	–	–	0.01 (264.83)	1.01 (1.51)
% Negative marginal costs	6.96	1.30	6.30	1.04	0.68	0.06
Observations	5,302	5,302	5,302	5,302	5,302	5,302
Adjusted R-squared	0.90	0.82	0.91	0.83	–	–
Firm fixed-effects	No	No	No	No	Yes	Yes

Columns 1–4 report nested logit specifications estimated on monthly data. Columns 5–6 report the Random Coefficients Logit estimated via PyBLP with absorbed firm fixed effects. OI: Optimal Instruments. The symbols ***,**,* represent statistical significance at the 1%, 5%, and 10% levels, respectively. DDD: Defined Daily Dose.

Table 4 shows the results of the one-level nested logit (NL1), the two-level nested logit (NL2), their instrumented counterparts (NL1 IV and NL2 IV), and the Random Coefficients Logit (RC Logit) with and without optimal instruments (OI). The RCL specifications include firm fixed effects. σ_1 and σ_2 measure the correlation of preferences within each nest. α represents the price coefficient. σ_α, σ_B and σ_G represent the standard deviations of the random coefficients on price, a branded product indicator, and a generic product indicator, respectively. The first-stage regressions for the NL2 IV specification are reported in Appendix A.

In the NL1 specification without instruments, the price coefficient is negative and statistically significant, though small in magnitude ($\hat{\alpha} = -0.06$). The nesting parameter σ_2 is close to unity, suggesting very strong within-nest correlation in preferences across pharmaceutical forms. When moving to the NL2 model, both σ_1 and σ_2 are statistically significant, supporting a two-level nesting structure in which consumers first choose the pharmaceutical form and then select between branded and generic alternatives within that form. Both NL1 and NL2 without instruments yield very high adjusted R-squared values (0.90 and 0.91), but the nesting parameters close to unity suggest that part of the within-group correlation may reflect endogenous variation in nested shares rather than true preference heterogeneity.

The NL1 IV and NL2 IV columns address the endogeneity of prices and inner-group shares using the instrumental variables described above. In the NL2 IV specification, the

price coefficient increases substantially in magnitude ($\hat{\alpha} = -0.18$), indicating considerably higher price sensitivity once endogeneity is corrected. Both nesting parameters drop relative to the non-instrumented models ($\hat{\sigma}_1 = 0.85$, $\hat{\sigma}_2 = 0.61$) but remain statistically significant and consistent with random utility theory ($0 \leq \sigma_2 \leq \sigma_1 \leq 1$).

The last two columns report the Random Coefficients Logit, with and without optimal instruments (Reynaert and Verboven, 2014). The RC Logit with standard instruments yields a negative and significant price coefficient ($\hat{\alpha} = -0.60$, s.e. $= 0.09$), but the standard deviations of the random coefficients are not identified: σ_α, σ_B and σ_G are estimated at or near zero with very large standard errors. With optimal instruments, the price coefficient becomes larger in magnitude ($\hat{\alpha} = -0.85$, s.e. $= 0.11$) and the standard deviation on price ($\sigma_\alpha = 0.15$) is also significant, indicating meaningful heterogeneity in price sensitivity. However, the random coefficients on the branded and generic indicators remain insignificant ($\sigma_B = 0.00$, $\sigma_G = 1.01$ with s.e. $= 1.51$), confirming the weak identification concern discussed earlier: in a single national market with only temporal variation, the data do not provide sufficient information to precisely estimate heterogeneity in preferences over these product attributes. This supports the choice of the two-level nested logit as the primary specification, where the nesting structure captures this form-level and brand/generic heterogeneity through σ_1 and σ_2.

6.2 Substitution patterns: consumers switch within form, not across forms

To examine substitution patterns, price elasticities are computed using the NL2 IV specification. For each market, I compute the $J_t \times J_t$ matrix of own- and cross-price elasticities. From here on I refer to the mean values reported in Table 5.

Starting with the full sample, the mean own-price elasticity (OPE) is -12.60, indicating substantial price sensitivity in the TRT market. The cross-price elasticities (CPE) confirm the expected substitution patterns implied by the nested structure: the average CPE within the same subnest (i.e., among products sharing both pharmaceutical form and brand/generic status) is 1.58, substantially higher than the CPE within the same nest but different subnest , and essentially zero across nests. This hierarchical pattern means that when the price of a product increases, consumers are most likely to substitute toward the closest alternative sharing the same form and brand/generic status. For instance, a price increase in a branded transdermal gel packet primarily shifts demand toward other branded transdermal gel packets, rather than toward generic packets or products in other forms. The smaller but positive CPE within the same nest but across subnests indicates some substitution between branded and generic versions of the same form, while the near-zero CPE across nests confirms that substitution across pharmaceutical forms is negligible. These results support the validity of the two-level nesting structure.

When disaggregating by drug form, topical gel pump exhibits the highest price sensitivity, followed by patch/disc and transdermal gel packet. Topical solution pump is the least elastic form. The within-subnest CPEs are highest for patch/disc, which is consistent with the fact that this form contains only branded products competing closely with one another. Notably, patch/disc has no CPE within the nest (reported as –) because no generic alternatives exist in this form, meaning the lower-level subnest coincides with the nest itself. The topical gel pump and topical solution pump forms show the highest CPE across subnests, indicating that in these forms there is meaningful substitution between branded and generic alternatives.

When distinguishing between branded and generic products, own-price elasticities are similar in magnitude. However, the within-subnest CPE is notably higher for branded products than for generics, suggesting that branded products face stronger competitive pressure from close substitutes within their subnest. This is consistent with the fact that branded products in forms with available generics compete more intensely with other branded alternatives.

Overall, these findings confirm that substitution primarily occurs among products that share both pharmaceutical form and brand/generic status, with limited substitution between branded and generic versions of the same form, and negligible substitution across forms. The magnitude of the own-price elasticities is higher than those typically reported in the pharmaceutical demand estimation literature, where values range from approximately -1.5 to -3.5 (Dutta, 2011; Méndez, 2018). This difference is likely driven by the combination of the high nesting parameters ($\hat{\sigma}_1 = 0.85$, $\hat{\sigma}_2 = 0.61$), which amplify own-price effects through the nested logit structure. Despite the higher OPEs, the cross-price elasticities remain moderate, consistent with a market where products are differentiated along the form dimension and switching costs limiting substitution across forms.

The magnitude of these elasticities should be interpreted with caution. As discussed in footnote 9, the estimated elasticities reflect the joint outcome of patient preferences and prescriber choices, as well as institutional features such as formulary placement, automatic substitution rules, and prior authorization requirements. The near-zero cross-form elasticities are consistent with the fact that switching across pharmaceutical forms typically requires a new prescription and physician involvement. Moreover, the prices used in estimation (pharmacy acquisition costs) closely approximate the wholesale acquisition cost, which the FTC's economic expert in the AndroGel® litigation defined as the appropriate measure for capturing competitive effects[18].

[18]He characterized average weighted price as the market price charged by pharmaceutical companies, and opined that it is the best metric for measuring the impact on payors, pharmacy benefit managers, and pharmacies because it captures total payments involved. FTC v. AbbVie Inc., No. 14-cv-05151, Findings of Fact and Conclusions of Law, p. 59 n.19 (E.D. Pa. June 29, 2018).

Table 5: Product-Level Price Elasticities

Category	OPE	CPE Subnest	CPE Nest	CPE Other
Total	-12.60 (10.38)	1.58 (3.34)	0.16 (0.58)	0.00 (0.01)
Topical gel pump	-18.74 (12.63)	1.38 (2.89)	0.37 (1.04)	0.00 (0.00)
Transdermal gel packet	-9.63 (8.44)	1.62 (3.38)	0.06 (0.09)	0.00 (0.01)
Patch/Disc	-11.62 (7.18)	2.44 (4.70)	–	0.00 (0.00)
Topical solution pump	-7.88 (3.66)	1.18 (2.53)	0.37 (0.51)	0.00 (0.01)
Branded	-12.60 (10.93)	2.46 (4.67)	0.15 (0.36)	0.00 (0.00)
Generic	-12.60 (9.74)	1.06 (2.01)	0.16 (0.74)	0.00 (0.01)

Mean values and standard deviations (in parentheses) of the product-level own-price elasticities (OPE) and cross-price elasticities (CPE) based on the estimated parameters from specification NL2 IV of Equation (4) and formulas in the Appendix; 5,302 observations.

6.3 Counterfactual: earlier generic entry

Table 6 presents the mean prices by drug form and firm type, disaggregated by package size and strength, comparing the observed prices to those simulated under the counterfactual scenario in which the PFD agreement does not exist and generic entry occurs earlier.

Table 6: Mean original and simulated prices by form and presentation prior to actual generic entry (with number of observations)

Presentation	Branded		% Diff	Generic	Generic vs.
	original	simulated		simulated	branded (%)
Topical gel pump					
12.5 g (1%), 75 g	9.49 (149)	8.18 (149)	−13.85	4.10 (144)	−49.9
20.25 g (1.62%), 75 g	13.29 (46)	11.13 (46)	−16.30	–	–
12.5 g (1%), 150 g	9.92 (52)	9.92 (52)	0.00	–	–
10 mg (2%), 60 g	30.30 (47)	30.28 (47)	−0.07	–	–
Transdermal gel packet					
25 mg (1%), 2.5 g	8.24 (143)	8.21 (143)	−0.35	4.57 (216)	−44.4
50 mg (1%), 5 g	2.36 (245)	2.33 (245)	−1.14	1.33 (216)	−43.1
2.5 g (1.62%)	6.83 (27)	6.77 (27)	−0.95	–	–
1.25 g (1.62%)	16.05 (27)	15.98 (27)	−0.40	–	–
Patch/Disc (all)	6.30 (329)	6.30 (329)	−0.01	–	–
Topical solution pump (all)	7.12 (46)	7.12 (46)	−0.03	–	–

Mean original and simulated prices under the counterfactual scenario, for all markets prior to January 2015, by form and presentation (strength and package size). Prices are expressed in $/DDD.

The results directly speak to the core question of this paper. In the forms where

counterfactual generics enter the market the simulation produces a new, cheaper alternative for consumers, as well as pushing branded prices down. This suggests that consumers taking TRT from 2006 until 2015 paid a higher price due to the delay. In the topical gel pump form, the entry of a generic version of 12.5g (1 %), 75g pushes branded prices down by 14%. Not only that, but by entering the market, the generic firms would have priced their product 50% lower than their branded counterpart. This decline in branded prices is consistent with the competitive pressure exerted by generic entry, which erodes the branded manufacturer's market power in this segment.

In transdermal gel packet, the other form where AndroGel® is present, the story is similar. Even if the branded manufacturer reacts little, by decreasing its price around 0.4–1.1%; the generic alternative would be available at roughly 56–57% the price of the branded product.

In forms where no counterfactual generics are introduced (patch/disc and topical solution pump), branded prices remain essentially unchanged, with changes below 0.1%. These small movements do not reflect demand-side substitution across forms, the estimated cross-form elasticities are near zero (Table 5) but rather a mechanical consequence of the Bertrand-Nash equilibrium recalculation: the introduction of new products in other forms marginally alters the outside option share s_{0t}, which enters all firms' first-order conditions and produces negligible price adjustments even in unaffected segments. The magnitude of these effects confirms that the welfare impact of the PFD agreement is concentrated in the forms directly affected by the delayed entry.

Overall, the counterfactual simulation shows that the main channel of consumer benefit from earlier generic entry operates mainly through the expansion of the choice set rather than only through large branded price reductions. While branded prices decline meaningfully only in topical gel pump, the availability of cheaper generic alternatives in both topical gel pump and transdermal gel packet may directly benefit price-sensitive consumers. The welfare implications of these changes are captured by the consumer surplus analysis.

Table 7 compares the observed and simulated CS for each year, under the assumption that generic entry occurred earlier (i.e., in the absence of the PFD agreement), and calculates the PFD damages as in equation (14), or in other words, the welfare losses caused by delaying generic entry.

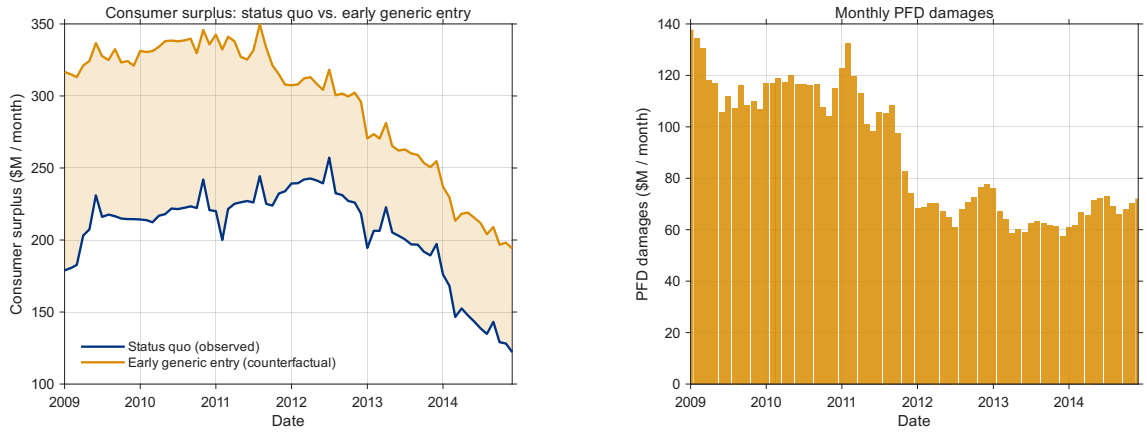
PFD damages are particularly high at the beginning of the panel. In 2009 the agreement cost consumers $1,405 million, so that consumer surplus was 36% below the level it would have reached under early generic entry. On average, the agreement cost consumers $89.6 million per month, over the whole delay period observed in the data (January 2009 to December 2014), total PFD damages amount to $6,451 million, equivalent to a 43.6% increase in consumer surplus relative to the status quo.

Figure 3 shows the two series month by month. The gap between the counterfactual

Table 7: Observed and simulated consumer surplus prior to actual generic entry

Period	CS Status quo	CS early entry	PFD Damages	in %
2009	2,460.3	3,865.3	1,405.0	57.11
2010	2,688.1	4,058.8	1,370.6	50.99
2011	2,708.2	3,972.5	1,264.3	46.69
2012	2,817.8	3,658.0	840.3	29.82
2013	2,393.7	3,148.4	754.7	31.53
2014	1,736.2	2,552.7	816.5	47.03
Monthly mean (2009m1–2014m12)	205.61	295.22	89.60	43.58
Total PFD Damages (2009m1–2014m12)	14,804.3	21,255.7	**6,451.4**	43.58

Observed (status quo) and simulated (early entry) consumer surplus computed as in equation (7) with the parameters from specification NL2 IV, for all markets prior to 1 January 2015. Expressed in millions $.



(a) Consumer surplus **(b)** PFD damages

Figure 3: Monthly consumer surplus under the status quo and under early generic entry, 2009m1–2014m12

Panel (a): consumer surplus computed as in equation (7) with the parameters from specification NL2 IV. Panel (b): monthly PFD damages. 3-month centred moving average.

and the observed consumer surplus opens immediately in January 2009 and never closes before actual generic entry. Monthly damages fluctuate between \$74 and \$138 million in 2009–2011 and between \$57 and \$78 million from 2012 onwards. Overall, the figure makes clear that the damage is persistent throughout the whole delay period.

Overall, the findings indicate that the PFD agreement's delay of generic entry had a particularly severe impact in the topical forms where AndroGel® is marketed. The total estimated PFD damages amount to \$6,451 million, or a 43.6% increase in consumer surplus relative to the status quo. These estimates should be interpreted as a lower bound of the true damages, for two reasons. First, due to data limitations, I do not observe the market prior to January 2009, but the delay period effectively started in 2006. Therefore, I am unable to account for the full extent of consumer harm during the initial three years of the delay. Second, the counterfactual introduces only the products of firms directly involved in the PFD agreement; additional generic entry that might have occurred in the absence of the settlement is not modeled.

Table 8: Variable profits by firm, product and presentation prior to actual generic entry: status quo vs. counterfactual (in millions \$)

Firm – product – presentation	**B/G**	**Status quo**	**Counterfactual**	**% Diff**
Topical gel pump		**2,273.3**	**2,545.0**	+12.0
AbbVie – AndroGel 1% – 12.5 g, 75 g	B	1,145.5 (72)	224.2 (72)	−80.4
AbbVie – AndroGel 1.62% – 20.25 g, 75 g	B	1,117.9 (44)	416.8 (44)	−62.7
Endo – Fortesta 2% – 10 mg, 60 g	B	9.8 (47)	0.6 (47)	−93.6
Perrigo/Paddock – generic AndroGel 1% – 12.5 g, 75 g	G*	–	1,291.8 (72)	–
Actavis – generic AndroGel 1% – 12.5 g, 75 g	G*	–	611.7 (72)	–
Transdermal gel packet		**2,960.3**	**2,778.5**	−6.1
AbbVie – AndroGel 1% – 50 mg, 5 g	B	1,913.2 (72)	1,671.6 (72)	−12.6
Endo/Auxilium – Testim 1% – 50 mg, 5 g	B	971.2 (72)	835.2 (72)	−14.0
AbbVie – AndroGel 1% – 25 mg, 2.5 g	B	61.4 (72)	53.7 (72)	−12.7
AbbVie – AndroGel 1.62% – 2.5 g	B	11.5 (27)	9.6 (27)	−17.1
AbbVie – AndroGel 1.62% – 1.25 g	B	2.7 (27)	2.3 (27)	−17.1
Perrigo/Paddock – generic AndroGel 1% – 25 mg, 2.5 g	G*	–	135.9 (72)	–
Actavis – generic AndroGel 1% – 25 mg, 2.5 g	G*	–	42.7 (72)	–
Par – generic AndroGel 1% – 25 mg, 2.5 g	G*	–	27.1 (72)	–
Perrigo/Paddock – generic AndroGel 1% – 50 mg, 5 g	G*	–	0.1 (72)	–
Actavis – generic AndroGel 1% – 50 mg, 5 g	G*	–	0.1 (72)	–
Par – generic AndroGel 1% – 50 mg, 5 g	G*	–	0.1 (72)	–

Total variable profits over 2009m1–2014m12, computed as in equation (12) with the parameters from specification NL2 IV, under observed prices and quantities (status quo) and under the counterfactual equilibrium with early generic entry. Expressed in millions \$. Number of monthly observations in parentheses. * denotes a fictitious generic product introduced in the counterfactual simulation.

Turning to the supply side of the market, Table 8 reports the total variable profits over the delay period (2009m1–2014m12) of the firms and products involved in the PFD agreement, under both the observed (status quo) and the simulated counterfactual scenario,

using Equation (12).

The first thing worth mentioning is the overall level of variable profits, in both forms. In the status quo, the variable profits for the topical gel pump form were $2,273 million, and $2,960 million in the transdermal gel packet form; under the counterfactual they become $2,545 million and $2,779 million, an increase of 12.0% in the former and a decrease of 6.1% in the latter. This indicates that generic entry not only pushed prices down, but also shared the profits with more players. The two forms differ, however, in what happens to the size of the pie. In topical gel pump, the two fictitious generics capture $1,903 million, more than the $1,623 million that AbbVie loses, so that early entry increases total variable profits in this form rather than merely redistributing them. In transdermal gel packet, by contrast, the $378 million lost by the branded 50 mg products of AbbVie and Endo is only partly recovered by the entrants ($206 million), which sell almost exclusively the 25 mg presentation, so that the pie shrinks.

Second, all branded products lose profits, by between 13% (AndroGel 1%, 50 mg packets) and 94% (Fortesta), as a result of the competitive pressure exerted by the generics. AndroGel's profits would have been about 80% lower in the 12.5 g pump presentation and about 63% lower in the 20.25 g pump presentation. The latter is AndroGel 1.62%, the reformulated product launched in 2011, which the FTC alleged AbbVie used to shift sales away from the formulation exposed to generic competition; according to the FTC, AbbVie sought a settlement that would give it "time to shift sales to its reformulated product, AndroGel 1.62%" (AbbVie Inc. v. FTC, 2020).

Comparing these counterfactual profits with those realized in the status quo shows why the settlement was profitable for AbbVie. Had generics entered in January 2009, AbbVie's variable profits over 2009m1–2014m12 would have been $2,378 million, against $4,252 million in the status quo. The difference must then be compared with what AbbVie paid for the delay. According to the FTC's complaint, as summarized by the Supreme Court, Solvay agreed to pay $12 million in total to Paddock, $60 million in total to Par, and an estimated $19–30 million per year, for nine years, to Actavis (Federal Trade Commission, 2013). Prorating these payments to the six years covered by the counterfactual and taking the upper bound for Actavis, AbbVie paid at most $252 million over 2009–2014, so that its profits net of payments were $4,000 million, still $1,622 million above the counterfactual. In other words, for every dollar paid to the generic challengers, AbbVie retained about seven dollars of profits.

Turning to the generics, the comparison between the payments received and the profits foregone is less uniform. Par received $60 million, more than twice the $27 million it would have earned by entering in 2009, so that the settlement was more profitable than entry and carried no litigation risk at all. One point deserves attention, which is the foregone profits for Actavis and Paddock. Actavis received between $114 and $180 million over the same six years, against $655 million of foregone profits, and Paddock received $12

million against $1,428 million of counterfactual profits, a gap that cannot be explained by litigation risk alone. There are things worth exploring more in depth regarding these players.

First, the Paddock/Par block. Since Actavis was the first one to file a Paragraph IV challenge, it had the key to block the market (Hemphill and Lemley, 2011). Moreover, Paddock and Par agreed to share the costs of the litigation in exchange for part of the potential profits from Paddock's (future) generic AndroGel product if that product gained FDA approval. This means that these players were unable to unblock the market and launch their product, because the 180-days of duopoly had not started yet. If they wanted to launch their product, they would have had to litigate their own case to a final judgment holding the #894 patent invalid or not infringed, a costly and uncertain route. Therefore, even if it seems that Paddock would have gotten more profits had it entered the market in Table 8, this does not take into account the potential litigation costs of suing their own case to a final judgement, and winning.

Second, this ex-post analysis might appear to show that Actavis left money on the table, by not exercising its right of entering into the market. Actavis's position was different, because as the first paragraph IV filer it held the 180-day exclusivity that, under the pre-2003 rules governing its application, did not lapse if left unused. By paying Actavis to stay out of the market, Solvay therefore purchased the delay of all three challengers at once, which explains why the bulk of the payments went to Actavis. For Actavis, the relevant comparison was not $655 million against $114–180 million, but a certain stream of payments, tied to AndroGel's profits and projected in 2006, against an uncertain prize net of litigation costs. With the figures in Table 8, accepting the settlement was rational for Actavis if it assigned a probability of about 28% or less to invalidating the '894 patent. This is precisely the inference the Supreme Court draws in *FTC v. Actavis* (Federal Trade Commission, 2013). A large payment that cannot be explained by litigation costs or by the services nominally provided reveals that the patent holder feared losing and that the challenger doubted winning, and Edlin et al. (2015) show that the inference is stronger when, as here, the first filer's exclusivity blocks additional entrants. Seen ex post, with a market that grew far beyond the 2006 projections, Actavis left money on the table, but seen ex ante, the payment was a safe substitute for a gamble it did not expect to win at that time.

7 Conclusions

Pay-for-delay agreements are legal in the United States unless a court finds, case by case, that their anticompetitive effects outweigh their justifications. This standard requires an estimate of the harm that one specific settlement caused to consumers in one specific market. This paper provides this estimate for the AndroGel® agreement, which is the

case that established the standard. Combining a two-level nested logit demand model, marginal costs recovered from Bertrand-Nash pricing, and a counterfactual in which the settling generics enter the market when they could have, I estimate that the delay cost consumers $6,451 million between January 2009 and December 2014 ($89.6 million per month), a 43.6% increase in consumer surplus that would have been realized under earlier entry. The loss is concentrated in the two forms in which AndroGel® was sold, and it operates mainly through the absence of a cheaper alternative rather than only through the branded price, which would have fallen by 14–16% in the gel pump and by 0.4–1.1% in the gel packet. The estimate should be interpreted as a lower bound, because the first three years of the delay are not observed and only the products of the settling firms are introduced.

The supply side of the model explains why the agreement was signed. Early entry would have reduced the branded manufacturer's variable profits over 2009–2014 from $4,252 million to $2,378 million. For every dollar that it paid to the generic challengers, it retained about seven dollars of profits, $1,622 million in total. For some of the challengers the agreement does not seem profitable ex post, but this comparison does not take into account litigation costs and the probability of winning or losing in court, which would likely increase the likelihood of entering into the agreement. Therefore, the agreement worked as a transfer of monopoly rent rather than as a source of efficiency.

Two features of the results matter for antitrust practice. First, the FTC's litigation over AndroGel® produced estimates of the brand's illicit profits, $1.2 billion according to the FTC's expert and $448 million according to the district court, but no estimate of consumer harm.[19] However, profits and consumer losses are different objects, and the rule of reason is concerned with the second one. The framework used here delivers it with product-level sales data and the entry dates and payments recorded in the settlement, inputs that are available in any litigated case. Second, because the harm arises mainly from the restricted choice set rather than from the branded price, assessments that focus on whether the brand cut its price after entry will understate the damage.

The analysis has some limitations, which point to further work. First, the model is static, so firms do not choose between entry, litigation and settlement over time. Embedding the demand system in a dynamic game would allow the timing of entry itself

[19]These figures do not refer to the 2006 settlement, which the FTC challenged separately in *FTC v. Actavis* and resolved by consent order in 2019 (Federal Trade Commission, 2019). They come from the FTC's 2014 suit over the patent infringement actions that AbbVie and Besins filed in 2011 against two later generic challengers, Teva and Perrigo (*FTC v. AbbVie Inc.*, 329 F. Supp. 3d 98, E.D. Pa. 2018). The FTC's expert, Carl Shapiro, estimated the defendants' illegal profits through August 2017 at $1.2 billion, assuming that absent the suits Teva would have launched by June 2012 and Perrigo by June 2013. The district court accepted the Perrigo counterfactual but not Teva's and reduced the figure to $448 million. Both numbers measure the difference between the brand's actual and but-for profits, not consumer surplus. The Third Circuit reversed the award, holding that Section 13(b) of the FTC Act does not authorize disgorgement (*FTC v. AbbVie Inc.*, 976 F.3d 327, 3d Cir. 2020), and the FTC withdrew its remaining claim in July 2021.

to be endogenous. Second, prices are pharmacy acquisition costs rather than patient out-of-pocket payments, so the welfare measure abstracts from copays, formulary tiers and reimbursement rules; claims data would improve it. Third, the elasticities reflect the joint decisions of patients, physicians and insurers rather than patient preferences alone. Finally, conditioning the damage estimate on the probability that the generic would have prevailed in court, the object that the theoretical literature identifies as central, would complete the link between this exercise and the rule-of-reason standard.

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A First stage regressions

Table 9: First Stage Regressions: Instruments and Covariates

	(1) σ_1	(2) σ_2	(3) Price
GH Local Rival Strength	0.005	0.010***	-0.318***
	(0.010)	(0.003)	(0.025)
GH Local Own Strength	0.526***	0.233***	1.291***
	(0.101)	(0.021)	(0.199)
GH Local Own (Strength, Size)	-0.464***	-0.339***	-3.163***
	(0.124)	(0.029)	(0.254)
Rival Sum Size	0.000	-0.001***	0.012***
	(0.000)	(0.000)	(0.001)
Own Sum Strength	0.095	-0.036**	1.343***
	(0.074)	(0.016)	(0.129)
Rival Mean Strength Nest	-2.075***	0.069**	1.795***
	(0.181)	(0.032)	(0.413)
Drug Size (DDD)	0.106***	0.002	0.035***
	(0.005)	(0.001)	(0.011)
Drug Strength (DDD)	0.194	0.346***	-17.852***
	(0.263)	(0.057)	(0.688)
Overall F-statistic	113.29	60.75	248.38
F-statistic (excluded instruments)	44.30	43.53	99.01
Sanderson-Windmeijer F-statistic	59.99	66.77	127.37

Notes: Each column corresponds to a separate first-stage regression for the three endogenous variables: σ_1 (within-subnest share), σ_2 (within-nest share), and price. Robust standard errors in parentheses. The F-statistic of excluded instruments tests the joint significance of the six excluded instruments in each first-stage equation. ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively.

A potential concern is that BLP-style instruments based on rival product counts may reflect the legally constrained market structure induced by the PFD agreement. However, the instruments are constructed from the observed competitive environment in each market period, which is the relevant object for identifying the price-share relationship: the demand model estimates how consumers respond to prices given the products actually available, regardless of why those products are available. The exclusion restriction requires that rival product characteristics affect demand only through their impact on prices and within-nest shares, not that the market structure itself be exogenous to the settlement. This requirement is plausible, since the product characteristics used to construct the instruments (drug strength and package size) are fixed at regulatory approval. The inclusion of the Gandhi and Houde (2019) differentiation instruments, which measure

proximity in the characteristic space rather than counting competitors, further reinforces this argument. The Sanderson-Windmeijer F-statistics (59.99, 66.77, and 127.37 for σ_1, σ_2, and price, respectively) confirm individual instrument relevance. The Hansen J statistic rejects at conventional levels ($p < 0.01$); this is common in demand estimation with large samples, where the test detects minor misspecification in the parametric model rather than instrument invalidity. I therefore rely primarily on the first-stage diagnostics and the economic plausibility of the exclusion restrictions.

B Simulation Algorithm

This appendix describes the fixed-point iteration algorithm used to compute Bertrand-Nash equilibrium prices under the counterfactual scenario. The simulation uses the structural parameters estimated from the NL2 IV specification and the multi-product firm pricing framework.

Step 1: Augment the product set

For each market t prior to January 2015, I introduce the eight generic products listed in Table 3. Since these products were not observed in the market prior to generic entry, their initial prices must be set externally. I initialize each fictitious product's price at 60% of the mean branded price in the same market and form. This is guided by empirical evidence showing that, upon actual entry, generic prices are approximately 60% of branded prices in the same form (Figure 4 and Figure 5).[20]

For existing products, the demand unobservable ξ_{jt} is set equal to the estimated residual from the NL2 IV demand equation. For fictitious products, ξ_{jt} is set equal to the mean residual of observed generic products in the post-entry period (markets $t \geq 73$), reflecting the average unobserved quality of generic products once they are actually marketed.

Marginal costs of the fictitious products are set as described in section 5: the recovered marginal cost of the AndroGel® product with the same form, strength and package size in month t, multiplied by the median generic-to-brand cost ratio observed for the corresponding real generic over 2017–2023 (between 0.26 and 0.69 depending on the presentation). The median is used because monthly recovered costs are noisy and occasionally negative; the mean ratios differ from the medians by at most 0.06.

Step 2: Compute mean utility and market shares

Given current prices $p^{(n)}$, the mean utility of product j in market t is:

$$\hat{\delta}_{jt} = \hat{\alpha} \cdot p_{jt}^{(n)} + \hat{\beta}_{\text{strength}} \cdot \text{strength}_j + \hat{\beta}_{\text{size}} \cdot \text{size}_j + \hat{\gamma} + \hat{\xi}_{jt}$$

where $\hat{\gamma}$ is the estimated intercept. Market shares s_{jt} are then computed using the two-level nested logit share equations (25–31), along with the within-group shares $s_{j|hg,t}$ and $s_{h|g,t}$.

[20] The simulation results are robust to this assumption. When fictitious prices are initialized at 100% of the branded price, the resulting outcomes change only marginally.

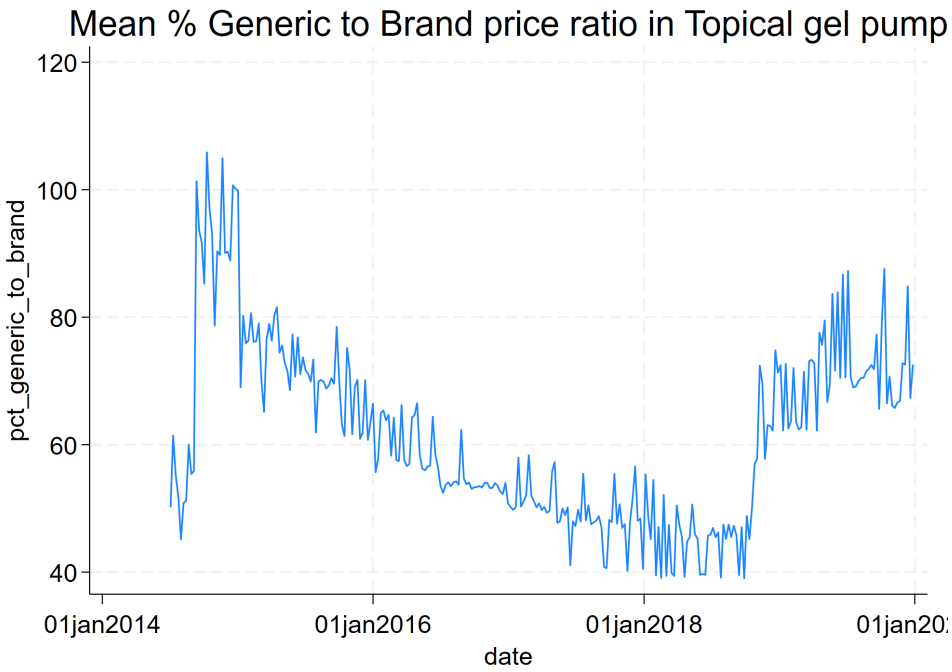


Figure 4: Mean % generic to branded price ratio for Topical Gel Pump form

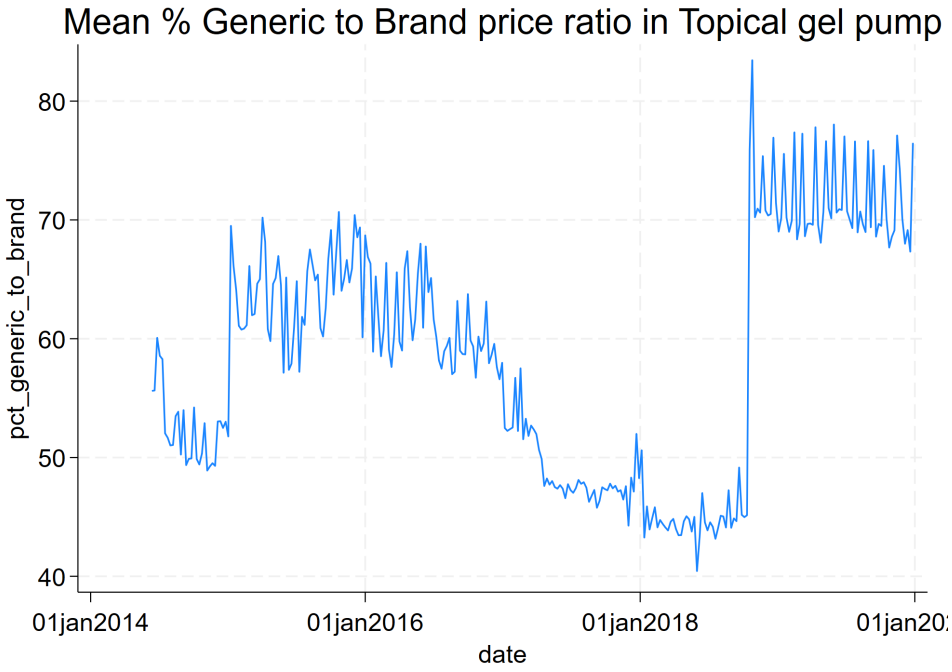


Figure 5: Mean % generic to branded price ratio for Transdermal Gel Packet form

Step 3: Compute markups and update prices

Prices are updated using the following equation derived from Equation 11:

$$p_j^{\text{new}} = c_j + \text{markup}_j$$

To improve numerical stability, prices are updated with a damping factor $\lambda = 0.5$:

$$p^{(n+1)} = \lambda \cdot p^{\text{new}} + (1 - \lambda) \cdot p^{(n)}$$

Step 4: Convergence

The algorithm iterates Steps 2–3 until convergence, defined by:

$$\max_j \left| p_j^{(n+1)} - p_j^{(n)} \right| < 10^{-6}$$

The algorithm converged for all markets in the dataset.

C The Hatch-Waxman Act

The Hatch-Waxman Act provided with mechanisms for generics to enter the market before patent expiration. In order to receive the Food and Drug Administration (FDA) approval, the generic must file an Abbreviated New Drug Application (ANDA), which must demonstrate *bioequivalency* of the branded drug. There are four ways of challenging the branded patent:

1. Paragraph I certification: The required patent information has not been filed
2. Paragraph II certification: The patent has already expired
3. Paragraph III certification: The patent will expire on entry date of the generic
4. Paragraph IV certification: The patent is either invalid or will not be infringed by the generic firm.

The generic manufacturer can submit a Paragraph IV certification to enter the market before the patent has expired. When doing so, it also triggers two additional provisions in the Hatch-Waxman act.

First, the generic firm must notify the branded firm of this patent litigation. This automatically triggers a period in which the branded firm has 45 days to file an infringement suit for patent challenge or it has the risk of facing immediate generic entry. Once the branded firm has filed a suit, this automatically triggers a 30 months stay to allow for litigation. During this 30 months period, the Food and Drug Administration (FDA) is prohibited to approve the generic until the court decision is made.

Second, the first generic firm to file a Paragraph IV certification is granted a 180-day exclusivity period, during which the FDA cannot approve other ANDAs; the period starts with the first filer's commercial launch. This means that no other generic can enter the market for 180 days since the date of entry of the first generic, generating a duopoly structure.

D Descriptive evidence of the TRT market

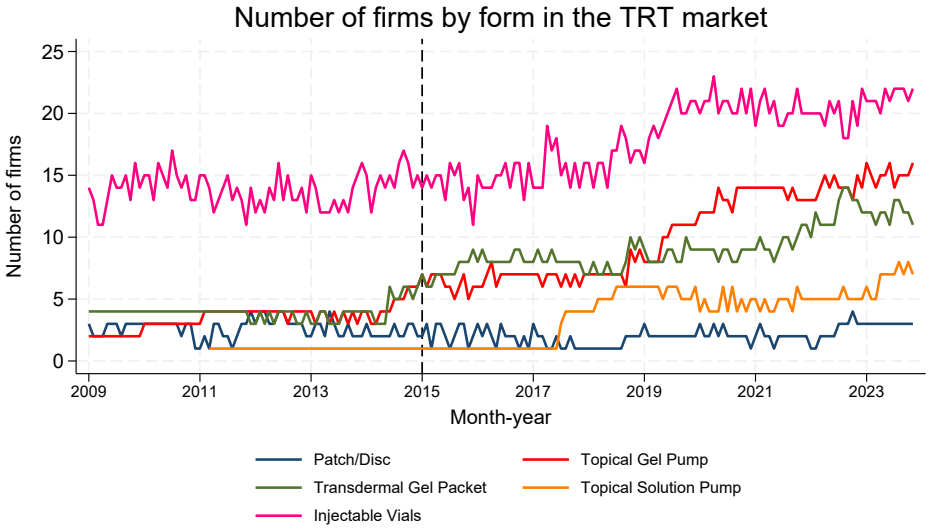


Figure 6: Number of firms by form. The black dashed line indicates first entry of generics in the market by Perrigo Pharmaceuticals.

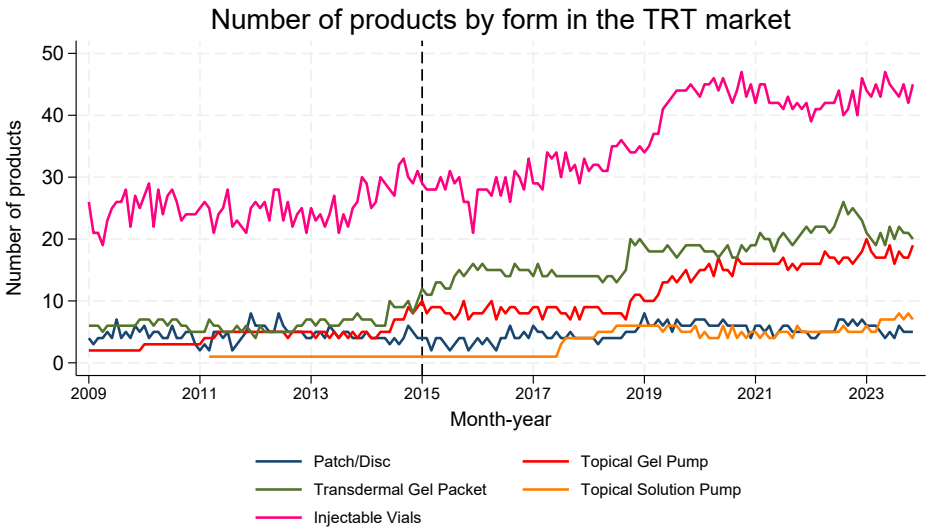


Figure 7: Number of products by form. The black dashed line indicates first entry of generics in the market by Perrigo Pharmaceuticals.

Table 10: Market structure by pharmaceutical form, pre- and post-generic entry

		Products			Firms			Median price	
Form	Period	Total	B	G	Total	B	G	Branded	Generic
Patch/Disc	Pre-entry	11	11	0	4	4	0	6.41	–
	Post-entry	13	13	0	5	5	0	17.00	–
Topical Gel Pump	Pre-entry	9	7	2	6	5	2	10.89	10.35
	Post-entry	23	8	15	17	6	13	23.03	14.94
Topical Solution Pump	Pre-entry	1	1	0	1	1	0	6.93	–
	Post-entry	9	1	8	9	1	8	12.25	10.76
Transdermal Gel Packet	Pre-entry	12	10	2	6	5	2	3.01	2.65
	Post-entry	30	10	20	14	5	11	10.24	6.46

Notes: Market structure before and after generic entry (January 2015). Products and firms are counted as unique identifiers observed in each period. B: branded; G: generic. Median prices are expressed in $/DDD. Pre-entry covers January 2009 to December 2014; post-entry covers January 2015 to November 2023. Missing values (–) indicate no generic products present in that form during the period.

E Random Coefficients Logit model

Utility Specification

Consumer i's conditional indirect utility for product j in market t is:

$$u_{ijt} = \alpha_i \cdot p_{jt} + \beta \cdot x_j + \gamma_f + \xi_j + \varepsilon_{ijt} \tag{15}$$

where p_{jt} is price in \$/DDD, $x_j = (\text{strength}_j, \text{size}_j)$ are observed product characteristics, γ_f are firm fixed effects, ξ_j is the demand unobservable, and $\varepsilon_{ijt} \sim$ i.i.d. Type I Extreme Value. The mean utility is:

$$\delta_{jt} = \alpha \cdot p_{jt} + \beta_{\text{strength}} \cdot \text{strength}_j + \beta_{\text{size}} \cdot \text{size}_j + \xi_j \tag{16}$$

Random Coefficients

Consumer heterogeneity enters through a non-linear utility component:

$$\mu_{ijt} = \sigma_\alpha \nu_{i\alpha} \cdot p_{jt} + \sigma_B \nu_{iB} \cdot \mathbf{1}[\text{branded}_j] + \sigma_G \nu_{iG} \cdot \mathbf{1}[\text{generic}_j] \tag{17}$$

where $\nu_{i\alpha}$, ν_{iB}, $\nu_{iG} \sim N(0,1)$ are independent standard normal draws, and $\sigma_\alpha \geq 0$, $\sigma_B \geq 0$, $\sigma_G \geq 0$ are the standard deviations of heterogeneity in price sensitivity, preference for branded products, and preference for generic products, respectively. The individual-level price coefficient is therefore $\alpha_i = \alpha + \sigma_\alpha \nu_{i\alpha}$. The branded and generic indicator dummies capture consumer-level taste heterogeneity over brand/generic status within each pharmaceutical form, which in the main specification is captured by the nesting parameters σ_1 and σ_2.

Market Shares

The unconditional market share of product j in market t is:

$$s_{jt}(\delta_t, \Sigma) = \int \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{k=1}^{J_t} \exp(\delta_{kt} + \mu_{ikt})} \, dP(\nu) \tag{18}$$

where $\Sigma = \text{diag}(\sigma_\alpha, \sigma_B, \sigma_G)$ collects the non-linear parameters.

Estimation

The model is estimated by GMM using the same instrument set as the NL2 IV specification. To address weak identification, the model is estimated in two stages following Reynaert and Verboven (2014): a first stage with standard instruments, followed by a second stage using approximate optimal instruments computed from the first-stage residuals. All estimations are performed using PyBLP (Conlon and Gortmaker, 2020).

F Simulation results from the Random Coefficients Logit model

As a robustness check, I replicate the counterfactual simulation described in Appendix B using the Random Coefficients Logit estimates. The simulation augments the product set with the same eight generic products listed in Table 3.

The counterfactual equilibrium is computed using PyBLP's routine[21] (Conlon and Gortmaker, 2020), which solves for Bertrand-Nash equilibrium prices via the Morrow and Skerlos (2011) fixed-point representation of the first-order conditions. This differs from the fixed-point iteration implemented in the NL2 IV simulation, which iterates directly on the price vector using Equation 10. Both approaches solve the same system of first-order conditions and converge to the same Bertrand-Nash equilibrium; the difference is purely algorithmic.

Table 11: Mean original and simulated prices by form and presentation prior to actual generic entry – RCL specification

Presentation	**Branded**		**% Diff**	**Generic**	**Generic vs.**
	original	simulated		simulated	branded (%)
Topical gel pump					
12.5 g (1%), 75 g	9.49 (149)	9.45 (149)	−0.44	5.25 (144)	−44.4
20.25 g (1.62%), 75 g	13.29 (46)	13.22 (46)	−0.54	–	–
12.5 g (1%), 150 g	9.92 (52)	9.90 (52)	−0.18	–	–
10 mg (2%), 60 g	30.30 (47)	30.26 (47)	−0.14	–	–
Transdermal gel packet					
25 mg (1%), 2.5 g	8.24 (143)	8.20 (143)	−0.49	4.97 (216)	−39.4
50 mg (1%), 5 g	2.36 (245)	2.34 (245)	−0.90	1.67 (216)	−28.3
2.5 g (1.62%)	6.83 (27)	6.79 (27)	−0.53	–	–
1.25 g (1.62%)	16.05 (27)	15.97 (27)	−0.49	–	–
Patch/Disc (all)	6.30 (329)	6.28 (329)	−0.26	–	–
Topical solution pump (all)	7.12 (46)	7.11 (46)	−0.22	–	–

Same structure as Table 6, using the RC Logit (OI) estimates. Prices in $/DDD; number of observations in parentheses. Generic simulated refers to the fictitious generic products only. Marginal costs of the existing products are recovered from the RCL first-order conditions; those of the fictitious generics follow the same rule as in the main specification (contemporaneous cost of the AndroGel® product with the same presentation times the median post-entry generic-to-brand cost ratio, here computed with the RCL-recovered costs, ratios 0.20–0.59). The demand unobservable includes the estimated firm fixed effects.

The results in Table 11 are qualitatively consistent with those obtained under the NL2 IV specification. In both cases the fictitious generics enter at prices well below the brand:

[21] Using Simulation.replace endogenous

28–44% below the simulated branded price within the same presentation under the RCL, against 43–50% under the NL2 IV. The main difference lies in the response of branded prices. Under the RCL, branded prices fall by less than 1% in every presentation, whereas the NL2 IV predicts reductions of 14–16% in the topical gel pump. This is consistent with the weak identification of the random coefficients discussed above: the standard deviation of the random coefficient on the branded indicator lies at the zero boundary and that on the generic indicator (1.01) is imprecisely estimated (standard error 1.51), so that brand/generic taste heterogeneity is only weakly identified. Overall, the main channel of the counterfactual, the availability of a generic alternative at roughly half the branded price, is robust to the choice of demand specification, while the size of the branded price response depends on it.

G Robustness check for the simulation

As an internal validation exercise, I simulate the Bertrand-Nash equilibrium using the estimated structural parameters (2NL) for the post-entry period (January 2015 onward) and the same algorithm described in Appendix B, where generic products are observed in the data, and compare the model's predicted prices to those actually observed.

The model replicates observed prices with high accuracy: the algorithm reproduces observed post-entry prices to numerical precision (relative errors below 10^{-5} percent in every segment), confirming that the supply-side first-order conditions and the recovered marginal costs are internally consistent with the data.

H Additional derivations

Elasticities

The own-price elasticity is given by:

$$\frac{\partial q_{jt}}{\partial p_{jt}} \frac{p_{jt}}{q_{jt}} = -\alpha p_{jt} \left(\frac{1}{1-\sigma_1} - \left(\frac{1}{1-\sigma_1} - \frac{1}{1-\sigma_2} \right) s_{j|h_g,t} - \left(\frac{\sigma_2}{1-\sigma_2} \right) s_{j|g,t} - s_{jt} \right) \tag{19}$$

The cross-price elasticities for drugs in the same sub-nest (branded or generic indication), h_g, are defined as:

$$\frac{\partial q_{jt}}{\partial p_{kt}} \frac{p_{kt}}{q_{jt}} = -\alpha p_{jt} \left(- \left(\frac{1}{1-\sigma_1} - \frac{1}{1-\sigma_2} \right) s_{j|h_g,t} - \left(\frac{\sigma_2}{1-\sigma_2} \right) s_{j|g,t} - s_{jt} \right) \tag{20}$$

Similarly, the cross-price elasticities for drugs in the same nest (drug form), g, are given by:

$$\frac{\partial q_{jt}}{\partial p_{kt}} \frac{p_{kt}}{q_{jt}} = -\alpha p_{jt} \left(- \left(\frac{\sigma_2}{1-\sigma_2} \right) s_{j|g,t} - s_{jt} \right) \tag{21}$$

Finally, we derive the cross-price elasticities with all drugs outside the own drug form to be:

$$\frac{\partial q_{jt}}{\partial p_{kt}} \frac{p_{kt}}{q_{jt}} = \alpha p_{jt} s_{jt} \tag{22}$$

Mathematical derivation of the 2 level Nested logit model

Utility Function

Assume there are J products in the market. Each product $j \in J$. The utility function of the individual i consuming a product j in time/market t is:

$$u_{ijt} = \alpha \cdot p_{jt} + \beta \cdot x_j + \xi_j + \epsilon_{ijt} \tag{23}$$

where α is the disutility of income, p_{jt} is the price, x_{jt} are the product characteristics and ϵ_{ijt} is a random shock to utility, distributed as i.i.d. I assume that the ϵ_{ijt} error term follows the distribution assumption of the two-level Nested Logit model.

The mean-utility parameter is:

$$\delta_{jt} = \alpha \cdot p_{jt} + \beta \cdot x_j + \xi_j \tag{24}$$

Nesting structure

Each market t can be partitioned into different G groups that represent the form of the drug.[22] Furthermore, these G groups can be partitioned into H_g subgroups, representing the indicator of the drug (whether this is branded or generic). Each subgroup h contains J_{hg} products, therefore $\sum_{g=1}^{G} \sum_{h=1}^{H_g} J_{hg} = J$.

These are the inclusive values:

$$I_{hg} = (1 - \sigma_1) \log \sum_{j=1}^{J_{hg}} \exp\left(\frac{\delta_{jt}}{1 - \sigma_1}\right) \tag{25}$$

$$I_g = (1 - \sigma_2) \log \sum_{h=1}^{H_g} \exp\left(\frac{I_{hg}}{1 - \sigma_2}\right) \tag{26}$$

$$I = \log(1 + \sum_{g=1}^{G} \exp(I_g)) \tag{27}$$

Using these values the probability of choosing product j conditional on its subgroup being selected is:

$$P(s_{j|hg}) = \frac{\exp(\frac{\delta_j}{1-\sigma_1})}{\exp(\frac{I_{hg}}{1-\sigma_1})} \tag{28}$$

The probability that subgroup h is selected conditional on its group being selected is:

[22] With a bit of abuse of notation, from now on I omit the subscript t.

$$P(s_{h|g}) = \frac{\exp(\frac{I_{hg}}{1-\sigma_2})}{\exp(\frac{I_g}{1-\sigma_2})} \tag{29}$$

The probability that the group g is selected is:

$$P(s_g) = \frac{\exp(I_g)}{\exp(I)} \tag{30}$$

And finally, the unconditional probability of choosing a product j is:

$$P(s_j) = P(s_g)P(s_{h|g})P(s_{j|hg}) \tag{31}$$

Derivation of Equation 4

Equation 31 cannot be estimated, because is non linear in the parameters, and in ξ_j which is inside δ_j. Berry (1994) proposes a transformation to make the error term enter linearly in the equation, commonly known as *Berry Inversion*.

The outside good is defined as:

$$s_0 = \frac{1}{1 + \sum_{g=1}^{G} \exp(I_g)} = \frac{\exp(\delta_0)}{\exp(I)} = \frac{1}{\exp(I)} \tag{32}$$

Using Equation 31 and Equation 32, we can construct the ratio $\frac{s_j}{s_0}$ as:

$$\frac{s_j}{s_0} = \frac{\exp(I_g)}{\exp(I)} \cdot \frac{\exp(\frac{I_{hg}}{1-\sigma_2})}{\exp(\frac{I_g}{1-\sigma_2})} \cdot \frac{\exp(\frac{\delta_j}{1-\sigma_1})}{\exp(\frac{I_{hg}}{1-\sigma_1})} \cdot \exp(I) \tag{33}$$

simplifying:

$$\frac{s_j}{s_0} = \frac{\exp(\frac{\delta_j}{1-\sigma_1})}{\exp(\frac{I_{hg}}{1-\sigma_1})} \cdot \frac{\exp(\frac{I_{hg}}{1-\sigma_2})}{\exp(\frac{I_g}{1-\sigma_2})} \cdot \exp(I_g) \tag{34}$$

Applying logarithms to Equation 34 we have:

$$\ln(s_j) - \ln(s_0) = \frac{\delta_j}{1-\sigma_1} - \frac{I_{hg}}{1-\sigma_1} + \frac{I_{hg}}{1-\sigma_2} - \frac{I_g}{1-\sigma_2} + I_g \tag{35}$$

Now let's group the I_{hg} and I_g terms:

$$\ln(s_j) - \ln(s_0) = \frac{\delta_j}{1-\sigma_1} + I_{hg} \cdot (\frac{1}{1-\sigma_2} - \frac{1}{1-\sigma_1}) + I_g \cdot (1 - \frac{1}{1-\sigma_2}) \tag{36}$$

In Equation 36 we can further simplify terms:

$$\frac{1}{1-\sigma_2} - \frac{1}{1-\sigma_1} = \frac{(1-\sigma_1) - (1-\sigma_2)}{(1-\sigma_1)(1-\sigma_2)} = \frac{\sigma_2 - \sigma_1}{(1-\sigma_1)(1-\sigma_2)} \tag{37}$$

and

$$1 - \frac{1}{1-\sigma_2} = \frac{(1-\sigma_2)-1}{1-\sigma_2} = -\frac{\sigma_2}{1-\sigma_2} \tag{38}$$

Substitute (37) and (38) into Equation 36 yields:

$$\ln(s_j) - \ln(s_0) = \frac{\delta_j}{1-\sigma_1} + I_{hg} \cdot (\frac{\sigma_2 - \sigma_1}{(1-\sigma_1)(1-\sigma_2)}) - I_g \cdot (\frac{\sigma_2}{1-\sigma_2}) \tag{39}$$

Now we want to isolate δ_j, so we add and subtract it:

$$\frac{\delta_j}{1-\sigma_1} = \delta_j + \frac{\delta_j}{1-\sigma_1} - \delta_j = \delta_j + \frac{\delta_j - \delta_j(1-\sigma_1)}{1-\sigma_1} = \delta_j + \sigma_1 \cdot \frac{\delta_j}{1-\sigma_1} \tag{40}$$

Therefore, using Equation 40 into Equation 39 we have:

$$\ln(s_j) - \ln(s_0) = \delta_j + \sigma_1 \cdot \frac{\delta_j}{1-\sigma_1} + I_{hg} \cdot (\frac{\sigma_2 - \sigma_1}{(1-\sigma_1)(1-\sigma_2)}) - I_g \cdot (\frac{\sigma_2}{1-\sigma_2}) \tag{41}$$

Now recall Equation 28 and Equation 29. If we take logarithms, we have:

$$\ln(s_{j|hg}) = \frac{\delta_j}{1-\sigma_1} - \frac{I_{hg}}{1-\sigma_1} \tag{42}$$

$$\ln(s_{h|g}) = \frac{I_{hg}}{1-\sigma_2} - \frac{I_g}{1-\sigma_2} \tag{43}$$

If we isolate Equation 42 and Equation 43 we have:

$$\frac{\delta_j}{1-\sigma_1} = \ln(s_{j|hg}) + \frac{I_{hg}}{1-\sigma_1} \tag{44}$$

$$\frac{I_{hg}}{1-\sigma_2} = \ln(s_{h|g}) + \frac{I_g}{1-\sigma_2} \tag{45}$$

Substitute Equation 44 into Equation 41 and find:

$$\ln(s_j) - \ln(s_0) = \delta_j + \sigma_1 \cdot \ln(s_{j|hg}) + \sigma_1 \cdot \frac{I_{hg}}{1-\sigma_1} + I_{hg} \cdot (\frac{\sigma_2 - \sigma_1}{(1-\sigma_1)(1-\sigma_2)}) - I_g \cdot (\frac{\sigma_2}{1-\sigma_2}) \tag{46}$$

Now let's simplify again I_{hg} terms in Equation 46:

$$\sigma_1 \cdot \frac{I_{hg}}{1-\sigma_1} + I_{hg} \cdot \left(\frac{\sigma_2 - \sigma_1}{(1-\sigma_1)(1-\sigma_2)}\right) = \frac{\sigma_1 I_{hg}(1-\sigma_2) + (\sigma_2 - \sigma_1) I_{hg}}{(1-\sigma_1)(1-\sigma_2)} \tag{47}$$

$$= \frac{I_{hg}\left[\sigma_1(1-\sigma_2) + \sigma_2 - \sigma_1\right]}{(1-\sigma_1)(1-\sigma_2)} \tag{48}$$

$$= \frac{I_{hg}\left[\sigma_1 - \sigma_1\sigma_2 + \sigma_2 - \sigma_1\right]}{(1-\sigma_1)(1-\sigma_2)} \tag{49}$$

$$= \frac{I_{hg}\left[\sigma_2 - \sigma_1\sigma_2\right]}{(1-\sigma_1)(1-\sigma_2)} \tag{50}$$

$$= \frac{I_{hg} \cdot \sigma_2(1-\sigma_1)}{(1-\sigma_1)(1-\sigma_2)} \tag{51}$$

$$= \frac{\sigma_2 I_{hg}}{1-\sigma_2} \tag{52}$$

Now substitute Equation 52 into Equation 46 and find:

$$\ln(s_j) - \ln(s_0) = \delta_j + \sigma_1 \cdot \ln(s_{j|hg}) + \frac{\sigma_2 I_{hg}}{1-\sigma_2} - I_g \cdot \left(\frac{\sigma_2}{1-\sigma_2}\right) \tag{53}$$

Arrange:

$$\ln(s_j) - \ln(s_0) = \delta_j + \sigma_1 \cdot \ln(s_{j|hg}) + \sigma_2 \cdot \left(\frac{I_{hg}}{1-\sigma_2} - \frac{I_g}{1-\sigma_2}\right) \tag{54}$$

where the last term comes from Equation 43, therefore:

$$\ln(s_j) - \ln(s_0) = \delta_j + \sigma_1 \cdot \ln(s_{j|hg}) + \sigma_2 \cdot \ln(s_{h|g}) \tag{55}$$

Which is Equation 4. ■

Derivation of Own and Cross-Price Elasticities

In deriving elasticities, I will use the inclusive values defined in Equation 25, Equation 26 and Equation 27.

Now define

$$D_{hg} \equiv \sum_{j=1}^{J_{hg}} \exp(\frac{\delta_j}{1-\sigma_1}) \tag{56}$$

Therefore, Equation 25 is:

$$I_{hg} = (1-\sigma_1) \cdot \ln(D_{hg}) \tag{57}$$

If we isolate D_{hg} in Equation 57:

$$D_{hg} = \exp(\frac{I_{hg}}{1-\sigma_1}) \tag{58}$$

Using this in Equation 28 yields:

$$s_{j|hg} = \frac{\exp(\frac{\delta_j}{1-\sigma_1})}{D_{hg}} \tag{59}$$

Now recall Equation 26:

$$I_g = (1-\sigma_2) \log \sum_{h=1}^{H_g} \exp\left(\frac{I_{hg}}{1-\sigma_2}\right) \tag{60}$$

Use Equation 57 in Equation 60:

$$I_g = (1-\sigma_2) \ln \left[\sum_{h=1}^{H_g} \exp\left(\frac{(1-\sigma_1) \ln(D_{hg})}{1-\sigma_2}\right)\right] = (1-\sigma_2) \ln \sum_{h=1}^{H_g} D_{hg}^{\frac{1-\sigma_1}{1-\sigma_2}} \tag{61}$$

Now define

$$D_g \equiv \sum_{h=1}^{H_g} D_{hg}^{\frac{1-\sigma_1}{1-\sigma_2}} \tag{62}$$

Therefore we have:

$$I_g = (1-\sigma_2) \ln(D_g) \tag{63}$$

Now take Equation 57 and divide both sides by $1-\sigma_2$

$$\frac{I_{hg}}{1-\sigma_2} = \frac{(1-\sigma_1) \cdot \ln(D_{hg})}{1-\sigma_2} = \frac{1-\sigma_1}{1-\sigma_2} \ln(D_{hg}) \tag{64}$$

Now take the exponent:

$$\exp\left(\frac{I_{hg}}{1-\sigma_2}\right) = D_{hg}^{\frac{1-\sigma_1}{1-\sigma_2}} \tag{65}$$

and use the resulting expression in Equation 29:

$$s_{h|g} = \frac{D_{hg}^{\frac{1-\sigma_1}{1-\sigma_2}}}{D_g} \tag{66}$$

From Equation 63 I can isolate D_g and have:

$$\exp(I_g) = D_g^{1-\sigma_2} \tag{67}$$

Substitute this expression into Equation 27 and find:

$$I = \ln(1 + \sum_{g=1}^{G} D_g^{1-\sigma_2}) \tag{68}$$

Now define:

$$D \equiv 1 + \sum_{g=1}^{G} D_g^{1-\sigma_2} \tag{69}$$

Therefore, we can take Equation 30 and express it as:

$$s_g = \frac{D_g^{1-\sigma_2}}{D} \tag{70}$$

Now we are ready to start deriving the elasticities, we will use Equation 59, Equation 66 and Equation 70 and plug the values into Equation 31:

$$s_j = \frac{\exp\left(\frac{\delta_j}{1-\sigma_1}\right)}{D_{h_g}} \cdot \frac{D_{h_g}^{\frac{1-\sigma_1}{1-\sigma_2}}}{D_g} \cdot \frac{D_g^{1-\sigma_2}}{D} \tag{71}$$

$$s_j = \frac{\exp\left(\frac{\delta_j}{1-\sigma_1}\right) \cdot D_{h_g}^{\frac{1-\sigma_1}{1-\sigma_2}-1} \cdot D_g^{1-\sigma_2-1}}{D} \tag{72}$$

$$s_j = \frac{\exp\left(\frac{\delta_j}{1-\sigma_1}\right) \cdot D_{h_g}^{\frac{\sigma_1-\sigma_2}{1-\sigma_2}} \cdot D_g^{-\sigma_2}}{D} \tag{73}$$

Now take logarithms in both sides:

$$\ln s_j = \frac{\delta_j}{1-\sigma_1} + \frac{\sigma_1-\sigma_2}{1-\sigma_2} \ln D_{h_g} - \sigma_2 \ln D_g - \ln D \tag{74}$$

Own price elasticity

Let's take the derivative of Equation 74 with respect to p_j, term by term:

$$\frac{\partial \ln s_j}{\partial p_j} = \frac{1}{s_j} \frac{\partial s_j}{\partial p_j} \tag{75}$$

Term 1:

$$\frac{\partial}{\partial p_j} \left(\frac{\delta_j}{1 - \sigma_1} \right) = \frac{\alpha}{1 - \sigma_1} \tag{76}$$

Recall that $\frac{\partial \delta_j}{\partial p_j} = \alpha$.

Term 2: $\frac{\partial}{\partial p_j}\left(\frac{\sigma_1-\sigma_2}{1-\sigma_2}\ln D_{h_g}\right)$
We need $\frac{\partial D_{h_g}}{\partial p_j}$:

$$\frac{\partial D_{h_g}}{\partial p_j} = \frac{\partial}{\partial p_j}\sum_{j}^{J\in J_{h_g}} \exp\left(\frac{\delta_j}{1-\sigma_1}\right) = \frac{\alpha}{1-\sigma_1}\exp\left(\frac{\delta_j}{1-\sigma_1}\right) \tag{77}$$

Therefore:

$$\frac{\partial \ln D_{h_g}}{\partial p_j} = \frac{1}{D_{h_g}}\cdot\frac{\alpha}{1-\sigma_1}\exp\left(\frac{\delta_j}{1-\sigma_1}\right) = \frac{\alpha}{1-\sigma_1}s_{j|h_g} \tag{78}$$

And the derivative becomes:

$$\frac{\partial}{\partial p_j}\left(\frac{\sigma_1-\sigma_2}{1-\sigma_2}\ln D_{h_g}\right) = \frac{\sigma_1-\sigma_2}{1-\sigma_2}\cdot\frac{\alpha}{1-\sigma_1}s_{j|h_g} \tag{79}$$

Term 3: $\frac{\partial}{\partial p_j}\left(-\sigma_2 \ln D_g\right)$
We need $\frac{\partial D_g}{\partial p_j}$:

$$\frac{\partial D_g}{\partial p_j} = \frac{\partial}{\partial p_j}\sum_{h=1}^{H_g} D_h^{\frac{1-\sigma_1}{1-\sigma_2}} = \frac{1-\sigma_1}{1-\sigma_2}\sum_{h=1}^{H_g} D_{h_g}^{\frac{1-\sigma_1}{1-\sigma_2}-1}\cdot\frac{\partial D_{h_g}}{\partial p_j} \tag{80}$$

where $\frac{\partial D_{h_g}}{\partial p_j}$ is taken from Equation 77.

$$= \frac{1-\sigma_1}{1-\sigma_2}\sum_{h=1}^{H_g} D_{h_g}^{\frac{\sigma_1-\sigma_2}{1-\sigma_2}}\cdot\frac{\alpha}{1-\sigma_1}\exp\left(\frac{\delta_j}{1-\sigma_1}\right) \tag{81}$$

$$= \frac{\alpha}{1-\sigma_2}\sum_{h=1}^{H_g} D_{h_g}^{\frac{\sigma_2-\sigma_1}{1-\sigma_2}}\exp\left(\frac{\delta_j}{1-\sigma_1}\right) \tag{82}$$

Therefore:

$$\frac{\partial \ln D_g}{\partial p_j} = \frac{1}{D_g}\cdot\frac{\alpha}{1-\sigma_2}D_{h_g}^{\frac{\sigma_2-\sigma_1}{1-\sigma_2}}\exp\left(\frac{\delta_j}{1-\sigma_1}\right) \tag{83}$$

Recall that $s_{h|g} = \frac{D_{h_g}^{\frac{1-\sigma_1}{1-\sigma_2}}}{D_g}$ and $s_{j|h_g} = \frac{\exp(\delta_j/(1-\sigma_1))}{D_{h_g}}$:

$$\frac{\partial \ln D_g}{\partial p_j} = \frac{\alpha}{1-\sigma_2}\cdot\frac{D_{h_g}^{\frac{\sigma_2-\sigma_1}{1-\sigma_2}}}{D_g}\cdot\frac{\exp(\delta_j/(1-\sigma_1))}{D_{h_g}} = \frac{\alpha}{1-\sigma_2}s_{h|g}\cdot s_{j|h_g} \tag{84}$$

Simplifying:

$$\frac{\partial \ln D_g}{\partial p_j} = \frac{\alpha}{1-\sigma_2}\cdot\frac{D_{hg}^{\frac{1-\sigma_1}{1-\sigma_2}}}{D_g}\cdot\frac{\exp(\frac{\delta_j}{1-\sigma_1})}{D_{hg}} \tag{85}$$

Therefore:

$$\frac{\partial}{\partial p_j}\left(-\sigma_2\ln D_g\right) = -\frac{\alpha\sigma_2}{1-\sigma_2}s_{h|g}\cdot s_{j|h_g} \tag{86}$$

Term 4: $\frac{\partial}{\partial p_j}(-\ln D)$

$$\frac{\partial D}{\partial p_j} = (1-\sigma_2)D_g^{-\sigma_2}\frac{\partial D_g}{\partial p_j} = (1-\sigma_2)D_g^{-\sigma_2}\cdot\frac{\alpha}{1-\sigma_2}D_{h_g}^{\frac{\sigma_2-\sigma_1}{1-\sigma_2}}\exp\left(\frac{\delta_j}{1-\sigma_1}\right) \tag{87}$$

where $\frac{\partial D_g}{\partial p_j}$ is taken from Equation 82.

$$= \alpha D_g^{-\sigma_2}D_{h_g}^{\frac{\sigma_2-\sigma_1}{1-\sigma_2}}\exp\left(\frac{\delta_j}{1-\sigma_1}\right) \tag{88}$$

Therefore:

$$\frac{\partial \ln D}{\partial p_j} = \frac{\alpha}{D}D_g^{-\sigma_2}D_{h_g}^{\frac{\sigma_1-\sigma_2}{1-\sigma_2}}\exp\left(\frac{\delta_j}{1-\sigma_1}\right) \tag{89}$$

Recall that $s_j = \frac{\exp(\delta_j/(1-\sigma_1))\cdot D_{h_g}^{\frac{\sigma_1-\sigma_2}{1-\sigma_2}}\cdot D_g^{-\sigma_2}}{D}$:

$$\frac{\partial \ln D}{\partial p_j} = \alpha s_j \tag{90}$$

Therefore:

$$\frac{\partial}{\partial p_j}(-\ln D) = -\alpha s_j \tag{91}$$

Now we need to group all terms from Equation 76, Equation 80, Equation 86 and Equation 91.

Recall that $\frac{\partial \ln(s_j)}{\partial p_j} = \frac{1}{s_j}\cdot\frac{\partial s_j}{\partial p_j}$

$$\frac{1}{s_j}\frac{\partial s_j}{\partial p_j} = \frac{\alpha}{1-\sigma_1} + \frac{(\sigma_1-\sigma_2)\alpha}{(1-\sigma_2)(1-\sigma_1)}s_{j|h_g} - \frac{\alpha\sigma_2}{1-\sigma_2}s_{h|g}s_{j|h_g} - \alpha s_j \tag{92}$$

Factor α

$$\frac{1}{s_j}\frac{\partial s_j}{\partial p_j} = \alpha\left[\frac{1}{1-\sigma_1} + \frac{\sigma_1-\sigma_2}{(1-\sigma_1)(1-\sigma_2)}s_{j|h_g} - \frac{\sigma_2}{1-\sigma_2}s_{h|g}s_{j|h_g} - s_j\right] \tag{93}$$

Re-write the second term as:

$$\frac{\sigma_1-\sigma_2}{(1-\sigma_1)(1-\sigma_2)} = \frac{1}{1-\sigma_2} - \frac{1}{1-\sigma_1} \tag{94}$$

Substitute:

$$\frac{1}{s_j}\frac{\partial s_j}{\partial p_j} = \alpha\left[\frac{1}{1-\sigma_1} + \left(\frac{1}{1-\sigma_2} - \frac{1}{1-\sigma_1}\right)s_{j|h_g} - \frac{\sigma_2}{1-\sigma_2}s_{h|g}s_{j|h_g} - s_j\right] \tag{95}$$

$$= \alpha\left[\frac{1}{1-\sigma_1}(1-s_{j|h_g}) + \frac{s_{j|h_g}}{1-\sigma_2} - \frac{\sigma_2}{1-\sigma_2}s_{h|g}s_{j|h_g} - s_j\right] \tag{96}$$

$$= \alpha\left[\frac{1}{1-\sigma_1}(1-s_{j|h_g}) + \frac{s_{j|h_g}}{1-\sigma_2}(1-\sigma_2 s_{h|g}) - s_j\right] \tag{97}$$

To find the definition of elasticity, multiply by p_j on both sides:

$$\frac{p_j}{s_j}\frac{\partial s_j}{\partial p_j} = \alpha p_j \left[\frac{1}{1-\sigma_1}(1 - s_{j|h_g}) + \frac{s_{j|h_g}}{1-\sigma_2}(1 - \sigma_2 s_{h|g}) - s_j\right] \tag{98}$$

If we define:

$$s_{j|g} \equiv s_{j|hg} \cdot s_j \tag{99}$$

Then the final formula for the own-price elasticity is:

$$\frac{p_j}{s_j}\frac{\partial s_j}{\partial p_j} = \alpha p_j \left[\frac{1}{1-\sigma_1} - \frac{1}{1-\sigma_1}s_{j|h_g} + \frac{1}{1-\sigma_2}s_{j|h_g} - \frac{\sigma_2}{1-\sigma_2}s_{j|g} - s_j\right] \tag{100}$$

Cross price elasticities

CPE same sub-nest

Recall that we depart from Equation 74.

We need $\frac{\partial \ln s_j}{\partial p_k}$ where $k \neq j$ but $k \in h_g$.

Term 1: $\frac{\partial}{\partial p_k}\left(\frac{\delta_j}{1-\sigma_1}\right) = 0$ (because δ_j does not depend on p_k)

Term 2: $\frac{\partial}{\partial p_k}\left(\frac{\sigma_1-\sigma_2}{1-\sigma_2}\ln D_{h_g}\right)$

$$\frac{\partial D_{h_g}}{\partial p_k} = \frac{\alpha}{1-\sigma_1}\exp(\delta_k/(1-\sigma_1)) \tag{101}$$

$$\frac{\partial \ln D_{h_g}}{\partial p_k} = \frac{\alpha}{1-\sigma_1}\cdot\frac{\exp(\delta_k/(1-\sigma_1))}{D_{h_g}} = \frac{\alpha}{1-\sigma_1}s_{k|h_g} \tag{102}$$

where the second inequality holds because of Equation 78

Therefore:

$$\frac{\partial}{\partial p_k}\left(\frac{\sigma_1-\sigma_2}{1-\sigma_2}\ln D_{h_g}\right) = \frac{\sigma_1-\sigma_2}{1-\sigma_2}\cdot\frac{\alpha}{1-\sigma_1}s_{k|h_g} = \frac{(\sigma_1-\sigma_2)\alpha}{(1-\sigma_1)(1-\sigma_2)}s_{k|h_g} \tag{103}$$

Term 3: $\frac{\partial}{\partial p_k}(-\sigma_2 \ln D_g)$

$$\frac{\partial D_g}{\partial p_k} = \frac{1-\sigma_1}{1-\sigma_2}D_{h_g}^{\frac{\sigma_1-\sigma_2}{1-\sigma_2}}\cdot\frac{\alpha}{1-\sigma_1}\exp(\delta_k/(1-\sigma_1)) \tag{104}$$

$$\frac{\partial \ln D_g}{\partial p_k} = \frac{\alpha}{1-\sigma_2}\cdot\frac{D_{h_g}^{\frac{\sigma_1-\sigma_2}{1-\sigma_2}}\cdot\exp(\delta_k/(1-\sigma_1))}{D_{h_g}\cdot D_g} \tag{105}$$

As in Equation 85 to Equation 86:

$$= \frac{\alpha}{1-\sigma_2}\cdot s_{k|h_g}\cdot s_{h|g} \tag{106}$$

Therefore:

$$\frac{\partial}{\partial p_k}(-\sigma_2 \ln D_g) = -\sigma_2\cdot\frac{\alpha}{1-\sigma_2}s_{k|h_g}\cdot s_{h|g} = -\frac{\alpha\sigma_2}{1-\sigma_2}s_{k|h_g}\cdot s_{h|g} \tag{107}$$

Term 4: $\frac{\partial}{\partial p_k}(-\ln D)$

$$\frac{\partial \ln D}{\partial p_k} = \alpha\cdot s_k \tag{108}$$

Therefore

$$\frac{\partial}{\partial p_k}(-\ln D) = -\alpha s_k \tag{109}$$

Now sum all terms in Equation 103, Equation 107 and Equation 109.

$$\frac{\partial \ln s_j}{\partial p_k} = \alpha \left[\frac{\sigma_1 - \sigma_2}{(1-\sigma_1)(1-\sigma_2)} s_{k|h_g} - \frac{\sigma_2}{1-\sigma_2} s_{k|h_g} \cdot s_{h|g} - s_k \right] \tag{110}$$

$$\frac{\sigma_1 - \sigma_2}{(1-\sigma_1)(1-\sigma_2)} = \frac{1}{1-\sigma_2} - \frac{1}{1-\sigma_1} \tag{111}$$

Therefore:

$$\frac{\partial \ln s_j}{\partial p_k} = \alpha \left[\left(\frac{1}{1-\sigma_2} - \frac{1}{1-\sigma_1} \right) s_{k|h_g} - \frac{\sigma_2}{1-\sigma_2} s_{k|g} - s_k \right] \tag{112}$$

Therefore, the cross price elasticity in the same subnest:

$$\frac{\partial s_j}{\partial p_k} \frac{p_k}{s_j} = \alpha p_k \left[- \left(\frac{1}{1-\sigma_1} - \frac{1}{1-\sigma_2} \right) s_{k|h_g} - \frac{\sigma_2}{1-\sigma_2} s_{k|g} - s_k \right] \tag{113}$$

CPE same nest but different sub-nest

If product k is in group g but in a different sub-nest $h' \neq h$:

Term 1: $\frac{\partial}{\partial p_k}(-\sigma_2 \ln D_g)$

Here D_g does depend on p_k because $k \in g$. We have:

$$\frac{\partial D_g}{\partial p_k} = \frac{1-\sigma_1}{1-\sigma_2} D_{h'_g}^{\frac{\sigma_1-\sigma_2}{1-\sigma_2}} \cdot \frac{\alpha}{1-\sigma_1} \exp\left(\frac{\delta_k}{1-\sigma_1} \right) \tag{114}$$

As in Equation 106:

$$\frac{\partial \ln D_g}{\partial p_k} = \frac{\alpha}{1-\sigma_2} s_{k|h'_g} s_{h'|g} \tag{115}$$

Therefore,

$$-\sigma_2 \frac{\partial \ln D_g}{\partial p_k} = -\frac{\alpha \sigma_2}{1-\sigma_2} s_{k|h'_g} s_{h'|g} = -\frac{\alpha \sigma_2}{1-\sigma_2} s_{k|g} \tag{116}$$

Term 2: $\frac{\partial}{\partial p_k}(-\ln D) = -\alpha s_k$

Now let's sum the terms

$$\frac{\partial \ln s_j}{\partial p_k} = \alpha \left[-\frac{\sigma_2}{1-\sigma_2} s_{k|g} - s_k \right]$$

Therefore, the cross price elasticity in different nests but same subnests:

$$\frac{\partial s_j}{\partial p_k} \frac{p_k}{s_j} = \alpha p_k \left[-\frac{\sigma_2}{1-\sigma_2} s_{k|g} - s_k \right] \tag{117}$$

CPE different nest

This is simply $\frac{\partial}{\partial p_k}(-\ln(D)) = -\alpha s_k$.

Therefore the Cross Price elasticity in a different nest:

$$\frac{\partial s_j}{\partial p_k}\frac{p_k}{q_j} = -\alpha p_k s_k \tag{118}$$

Derivation of Equation 7.

I depart from Equation 1 and simplify:

$$u_{ijt} = \delta_{jt} + \epsilon_{ijt} \tag{119}$$

where ϵ_{ijt} follows the nested logit Generalized Extreme Value (GEV) distribution. (McFadden, 1978) shows that:

$$E[\max_j u_{ij}] = \ln\left(\sum_j \exp(V_j)\right) + C \tag{120}$$

where (V_j) is the systematic utility and C is the Euler constant.
For the two-level nested logit, the structure is:

$$E[\max_j u_{ij}] = I + C \tag{121}$$

where I is defined as in Equation 27.
Now recall Equation 25 and Equation 56. Therefore:

$$I_{hg} = (1 - \sigma_2) \cdot \ln(D_{hg}) \tag{122}$$

$$I_{hg} = (1 - \sigma_2) \cdot \ln(\sum_{j=1}^{J_{hg}} \exp(\frac{\delta_j}{1 - \sigma_1})) \tag{123}$$

Substitute this in Equation 26.

$$I_g = (1 - \sigma_2) \cdot \ln\left(\sum_{h=1}^{H_g} \exp\left(\frac{(1 - \sigma_1) \cdot \ln(D_{hg})}{1 - \sigma_2}\right)\right) \tag{124}$$

$$I_g = (1 - \sigma_2) \cdot \ln\left(\sum_{h=1}^{H_g} D_{hg}^{\frac{1-\sigma_1}{1-\sigma_2}}\right) \tag{125}$$

Recall Equation 62. Therefore:

$$I_g = (1 - \sigma_2) \ln(D_g) \tag{126}$$

Taking exponents:

$$\exp(I_g) = D_g^{1-\sigma_2} \tag{127}$$

Recall Equation 27 and substitute using Equation 127:

$$I = \ln\left(1 + \sum_{g=1}^{G} D_g^{1-\sigma_2}\right) \tag{128}$$

Recall Equation 69, therefore:

$$I = \ln(D) \tag{129}$$

By definition, the Consumer Surplus of individual i is the maximum expected utility, converted into monetary units (in this case, the units are $\$/DDDs$).

$$CS_i = \frac{1}{\alpha} \cdot \ln(D) = \frac{1}{\alpha} \cdot \ln\left(1 + \sum_{g=1}^{G} D_g^{1-\sigma_2}\right) \tag{130}$$

The aggregated Consumer Surplus is the same as the individual, multiplied by the potential market size M_t, therefore:

$$CS_t = \frac{M_t}{\alpha} \cdot \ln\left(1 + \sum_{g=1}^{G} D_g^{1-\sigma_2}\right) \tag{131}$$

and if we expand $D_{g,t}$ using Equation 62 we arrive to:

$$CS_t = \frac{M_t}{\alpha} \ln\left(1 + \sum_{g=1}^{G} \left(\sum_{h=1}^{H_g} D_{h_g,t}^{\frac{1-\sigma_1}{1-\sigma_2}}\right)^{1-\sigma_2}\right) \tag{132}$$

where

$$D_{h_g,t} = \sum_{j \in h_g} \exp\left(\frac{\hat{\delta}_{jt}}{1-\sigma_1}\right) \tag{133}$$

and $\hat{\delta}_{jt}$ are the fitted values from Equation 2.